MODULE 5

CONFIDENCE INTERVALS

OUTLINE

1. Quick review
2. Overview of confidence intervals
3. Constructing 95% confidence intervals
4. Properties of confidence intervals
5. Inference by eye
6. Study questions

OUTLINE

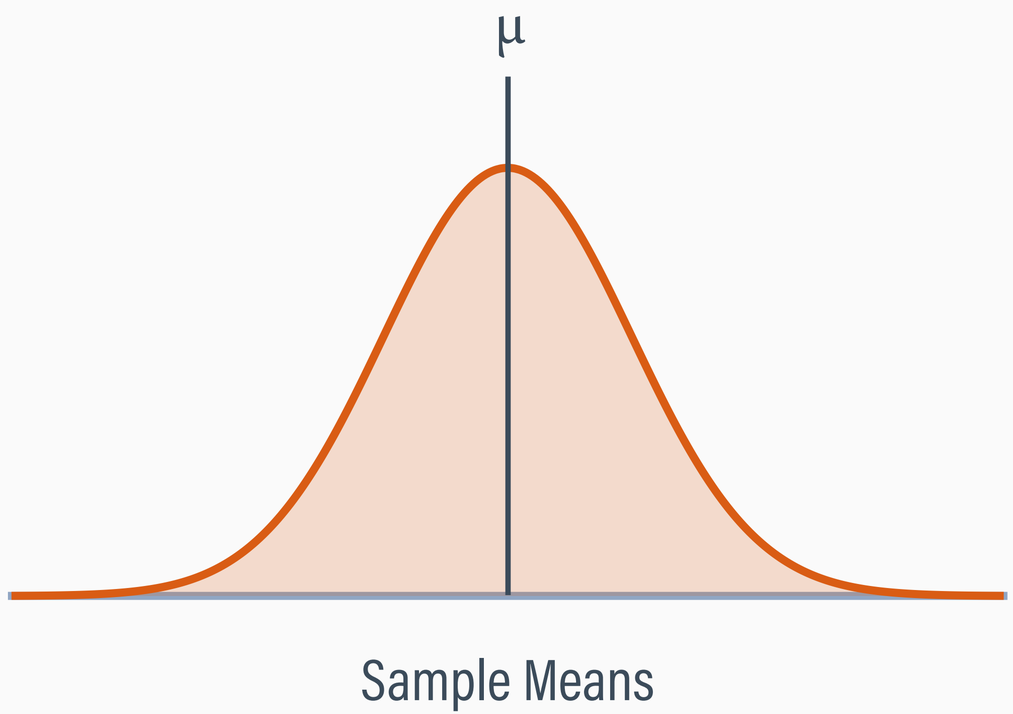
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QUICK REVIEW: SAMPLING ERROR

- The frequentist paradigm imagines a single population that spawns many hypothetical random samples of data (one parameter, many hypothetical estimates)
- The amount by which an estimate differs from the true population statistic is called sampling error
- Every hypothetical sample has a different amount of error

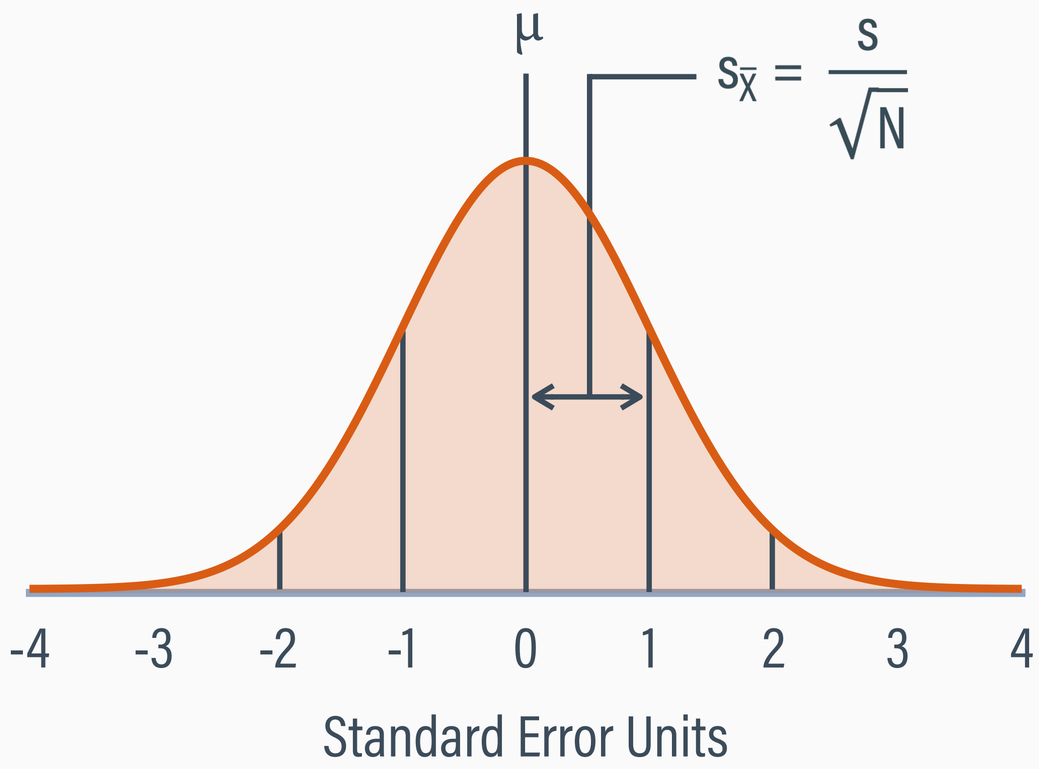
QUICK REVIEW: SAMPLING DISTRIBUTION

- The distribution of the estimates from many hypothetical samples is a sampling distribution
- With a large enough N, sample means follow a normal curve centered at the true mean
- Most estimates have small sampling errors, but a few have larger errors



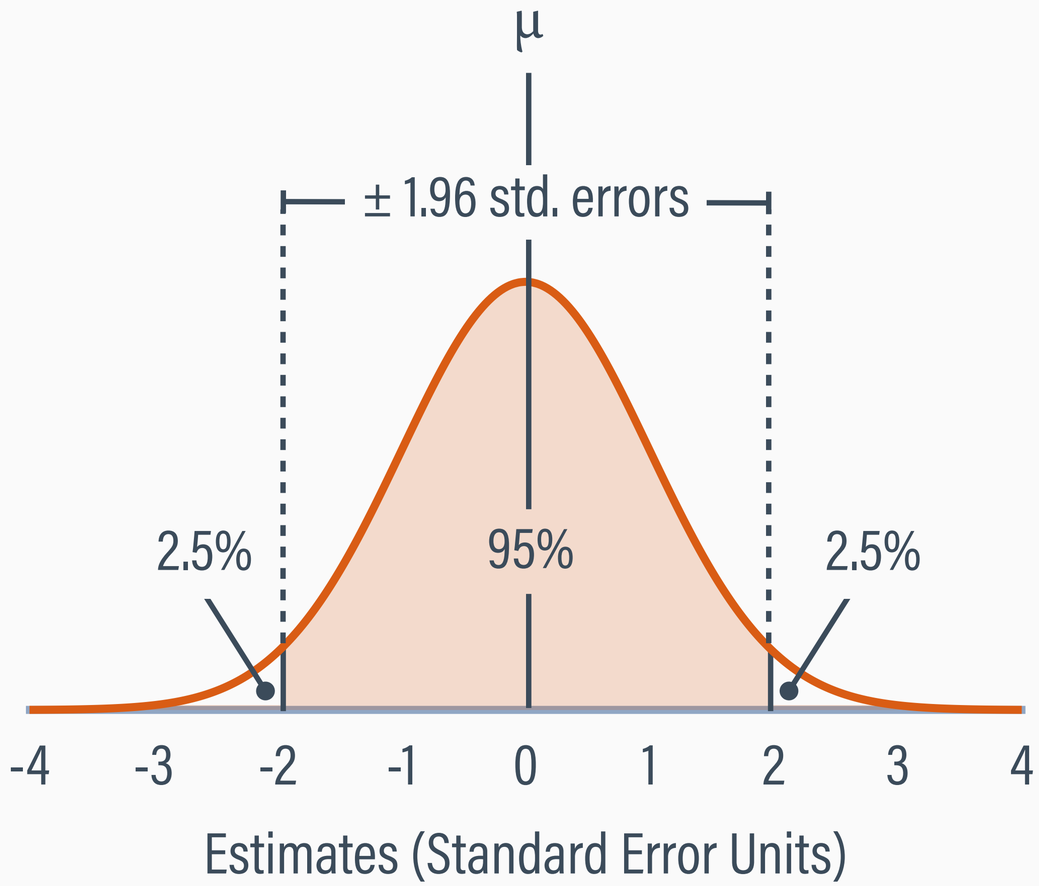
QUICK REVIEW: STANDARD ERROR

- The standard error is the average distance from a sample mean and the true mean
- $s_{\bar{X}}$ = standard deviation of the sample means
- The standard error is the average or expected amount of sampling error across many hypothetical samples



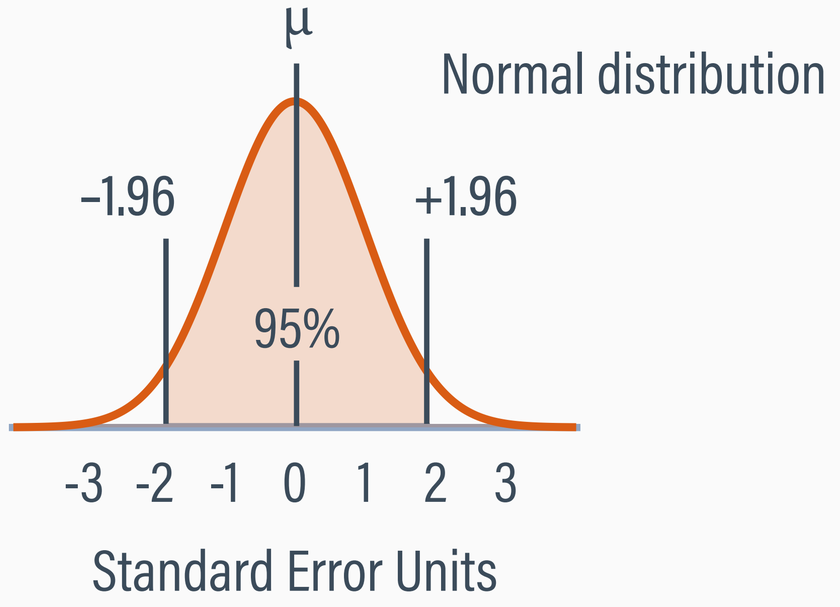
QUICK REVIEW: NORMAL CURVE RULE

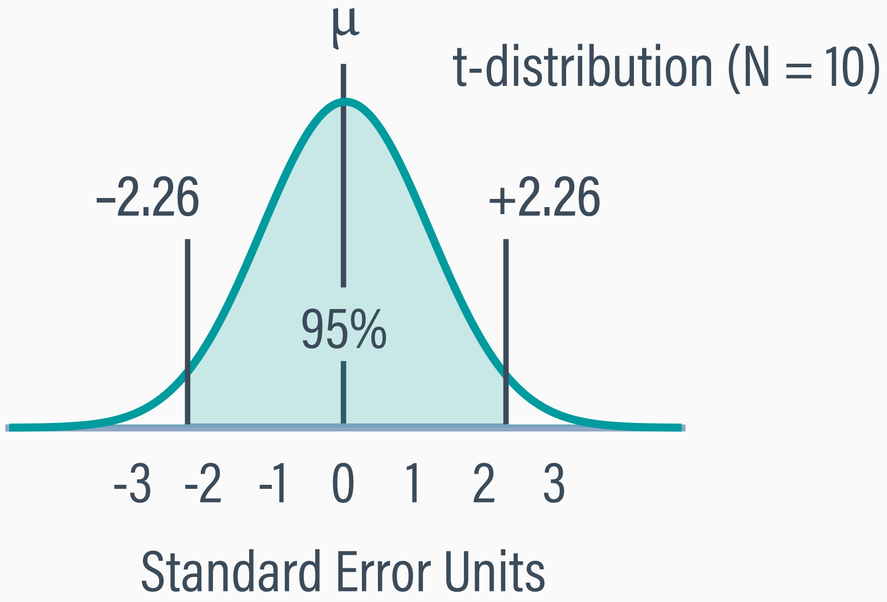
- The standard error is the standard deviation of many hypothetical sample means
- We can apply normal curve rules of thumb
- 95% of the means from large samples are within ± 1.96 standard errors of the true mean



QUICK REVIEW: T-DISTRIBUTION

- When using small samples, the normal curve is an inaccurate description of sampling error
- The t-distribution is a series of bell-shaped curves that stretch out (become more variable) as the N decreases
- Small samples are more likely to produce outlier estimates, and “stretching” the curve honors that





OUTLINE

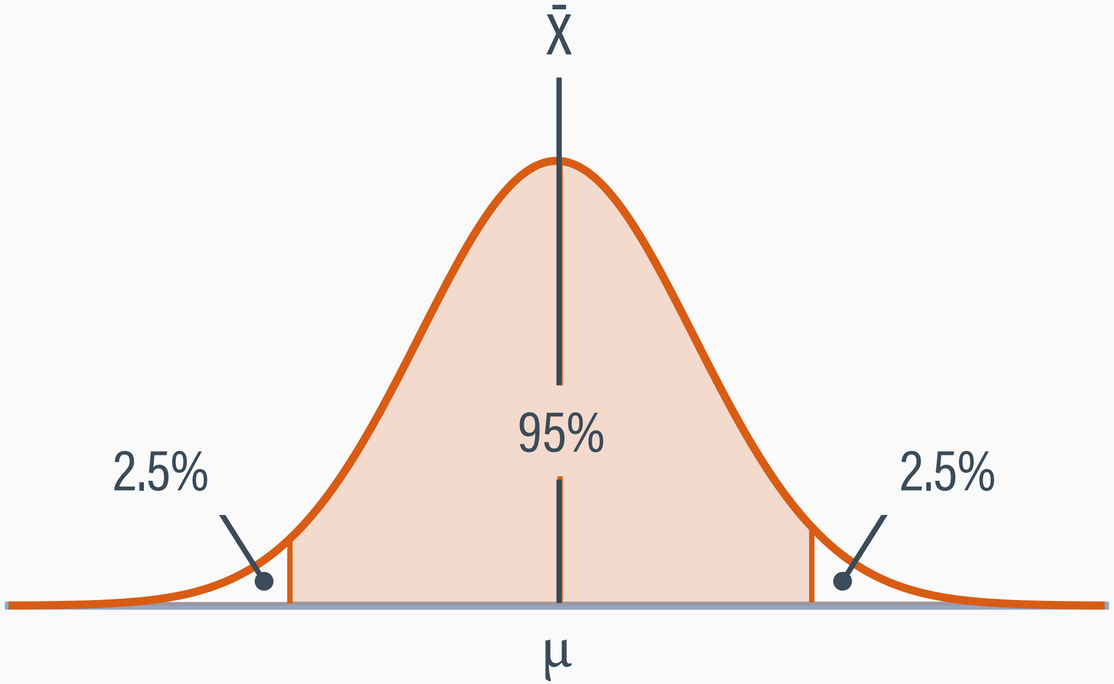
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CONFIDENCE INTERVAL OVERVIEW

- We have a single sample of data that gives us an estimate $\bar{X}$
- The sample mean could be exactly the same as the population mean ($\bar{X} = \mu$), too high ($\bar{X} > \mu$), or too low ($\bar{X} < \mu$)
- A confidence interval answers the following question: What are the lowest and highest values of μ that could have plausibly produced the $\bar{X}$ from our sample data?

MOST LIKELY POPULATION PARAMETER

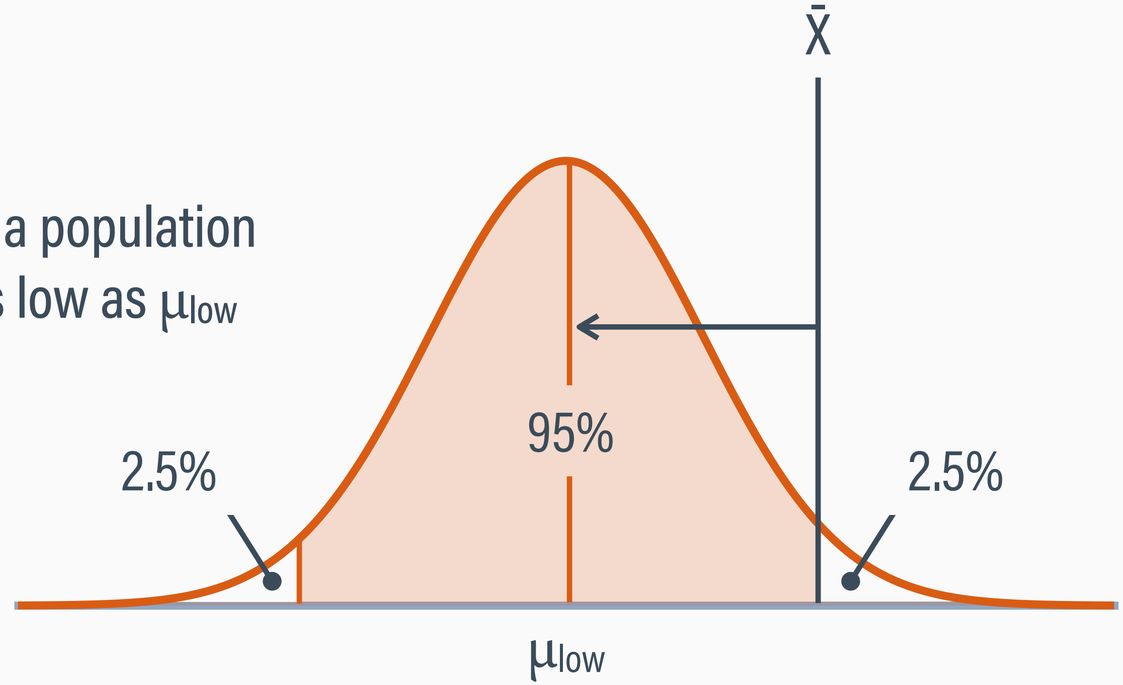
- The sample mean is our “best guess” about the true population mean, so $\bar{X}$ could sit exactly at the center of the sampling distribution ($\bar{X} = \mu$)



LOWEST PLAUSIBLE POPULATION MEAN

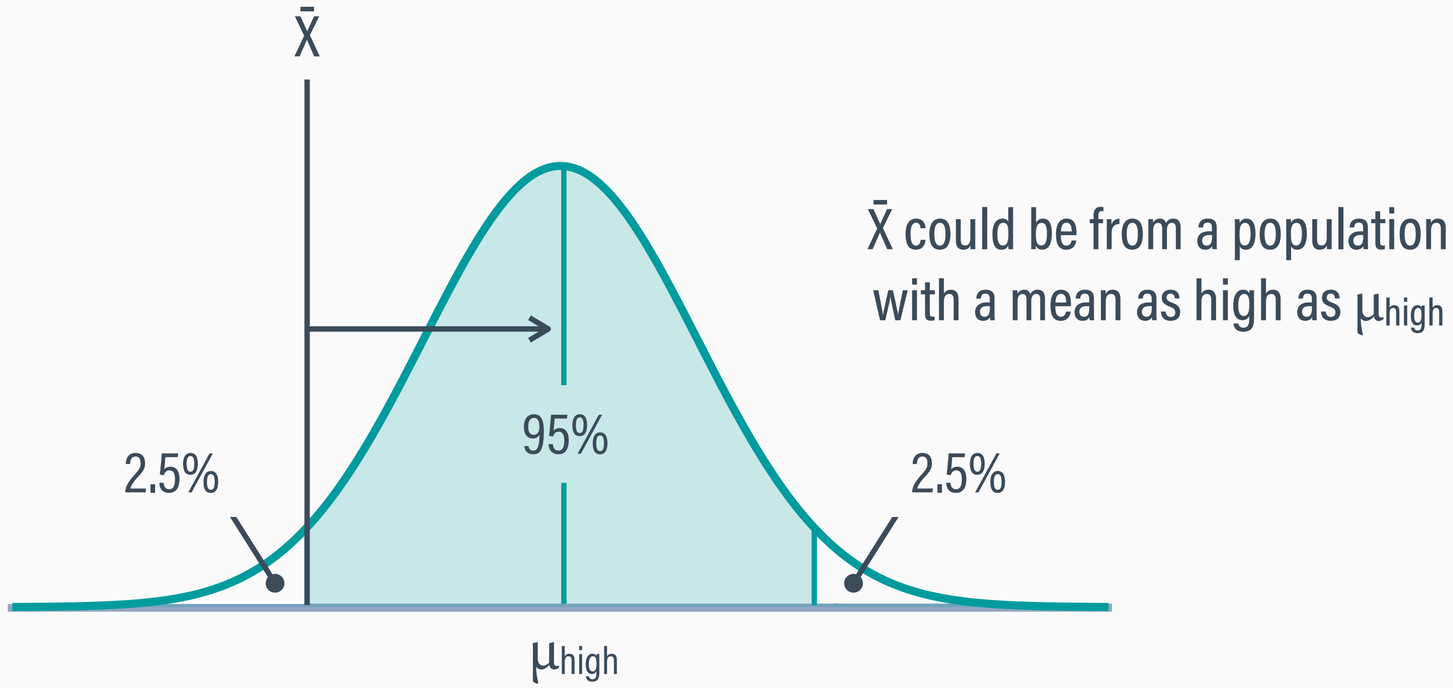
- Slide the sampling distribution down from $\bar{X}$ until the sample mean sits exactly at its 97.5th percentile—the center is the lowest population mean that could plausibly have produced our $\bar{X}$

$\bar{X}$ could be from a population with a mean as low as μ_{low}



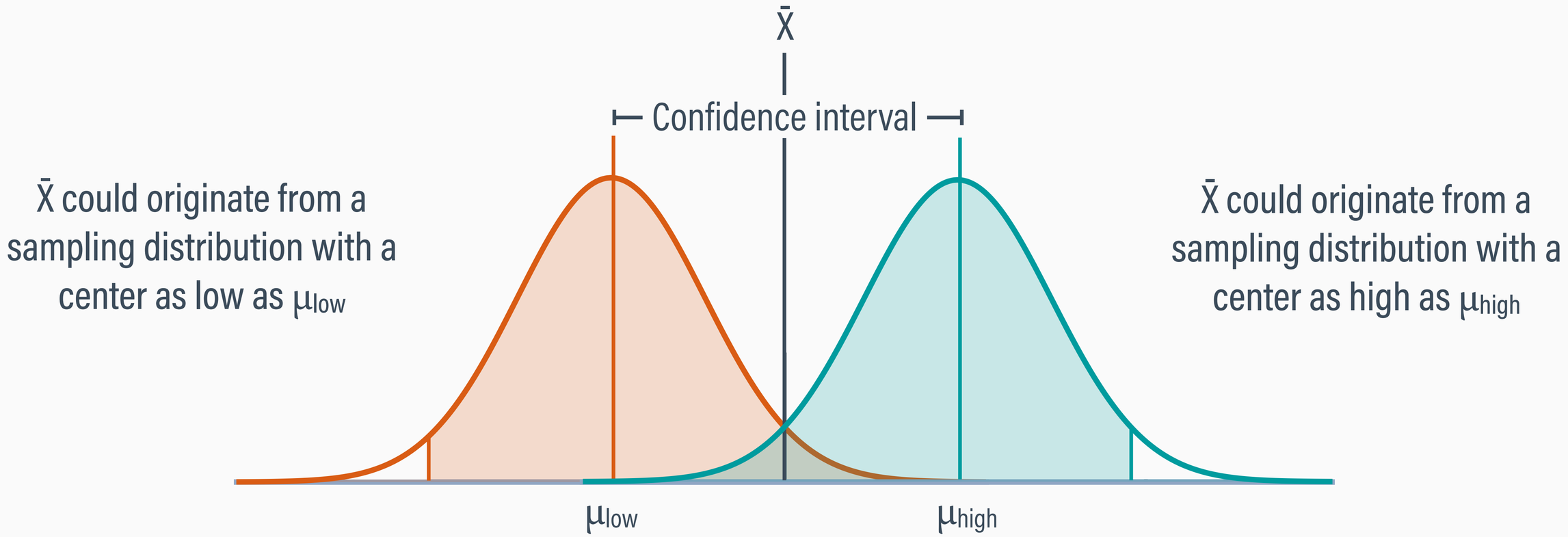
HIGHEST PLAUSIBLE POPULATION MEAN

- Slide the sampling distribution up from $\bar{X}$ until the sample mean sits exactly at its 2.5th percentile—the center is the highest population mean that could plausibly have produced our $\bar{X}$



EXTREME POPULATIONS INTERPRETATION

- A confidence interval gives the two most extreme values of the population mean that could have reasonably produced the data



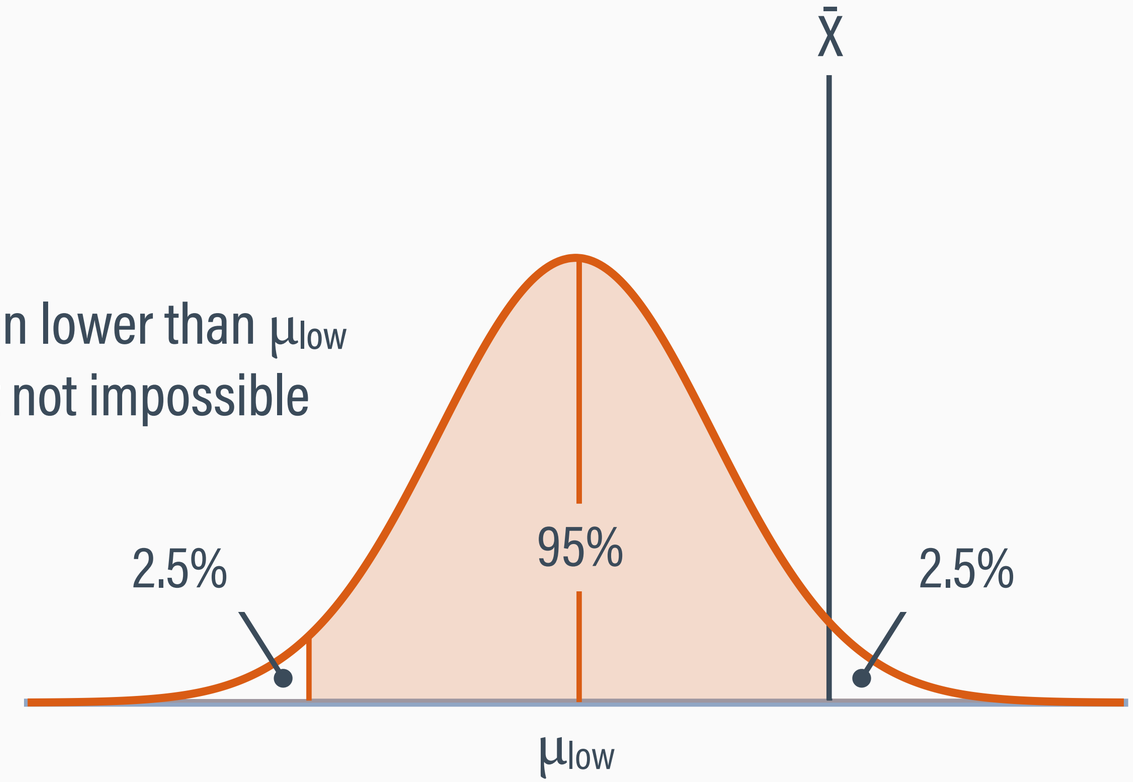
SETTING THE CONFIDENCE LEVEL

- An estimate should lie far (but not too far) from the lowest and highest plausible values of μ
- That distance is defined as a researcher-specified probability
- 95% confidence intervals predominate, but 90% and 99% are defensible and common in some applications

95% CONFIDENCE

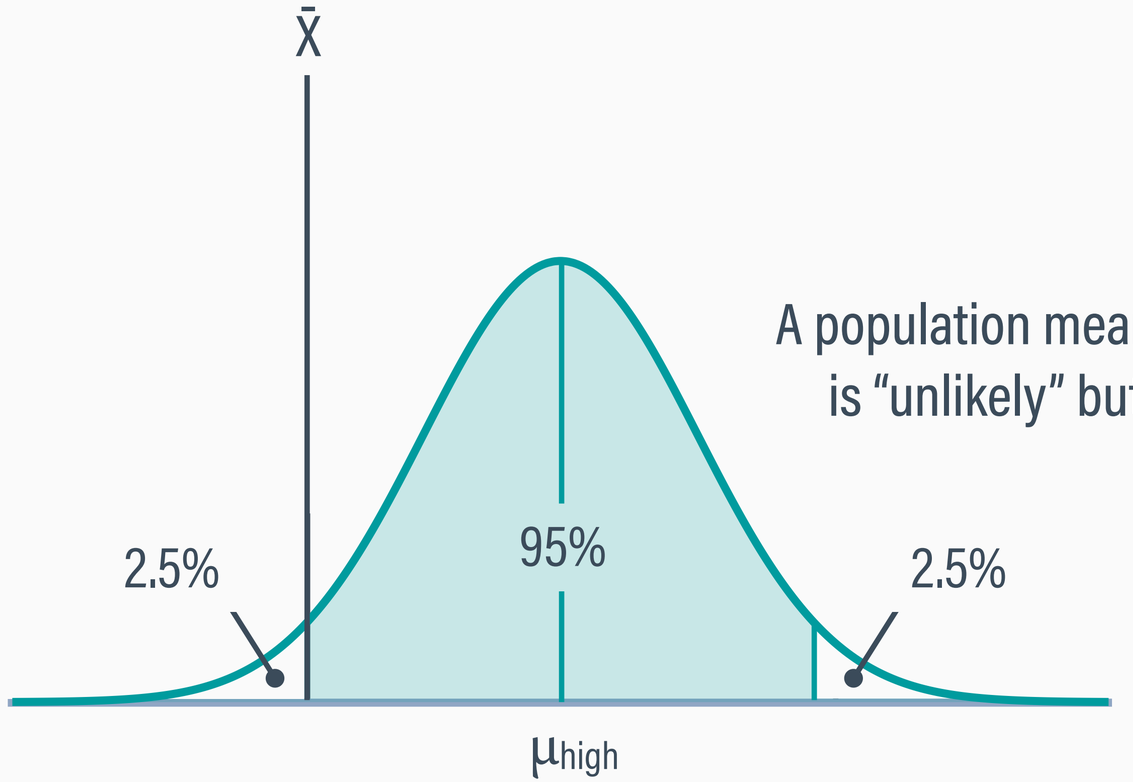
- 95% confidence implies that $\bar{X}$ sits at the upper boundary of the middle 95% (the 97.5th percentile) of all samples from a population with μ_{low}

A population mean lower than μ_{low} is “unlikely” but not impossible



95% CONFIDENCE

- 95% confidence implies that $\bar{X}$ sits at the lower boundary of the middle 95% (the 2.5th percentile) of all samples from a population with μ_{high}

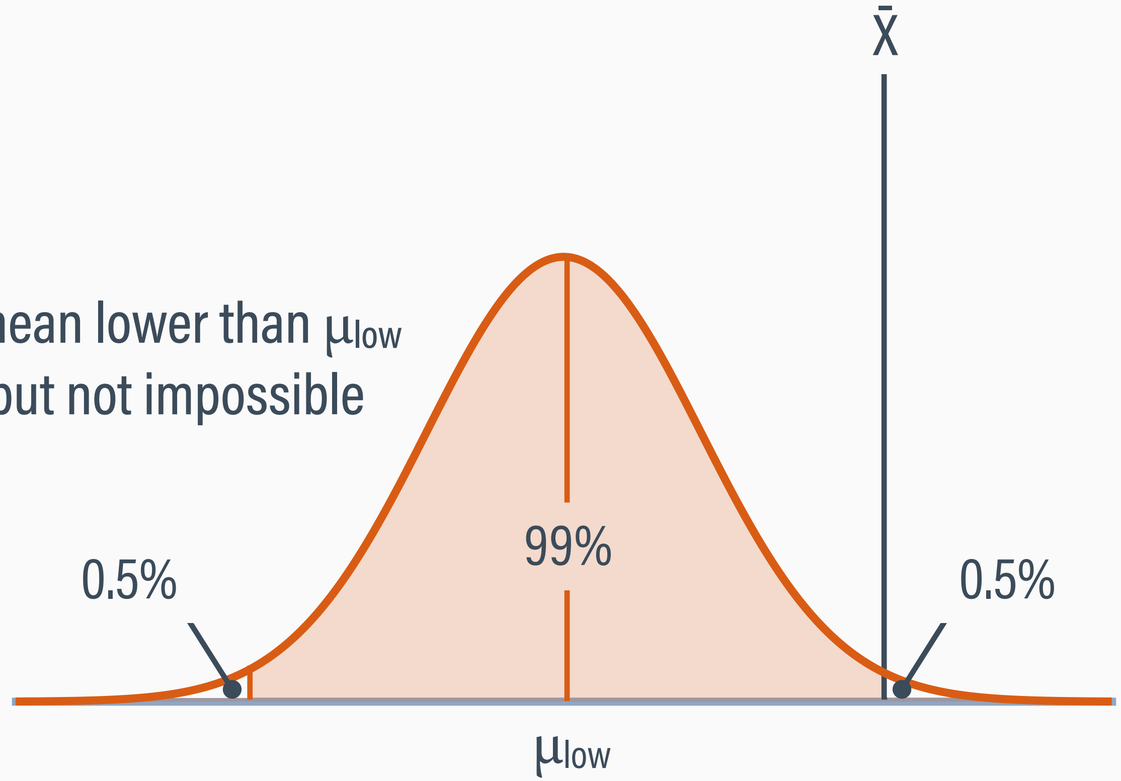


A population mean higher than μ_{high} is “unlikely” but not impossible

99% CONFIDENCE

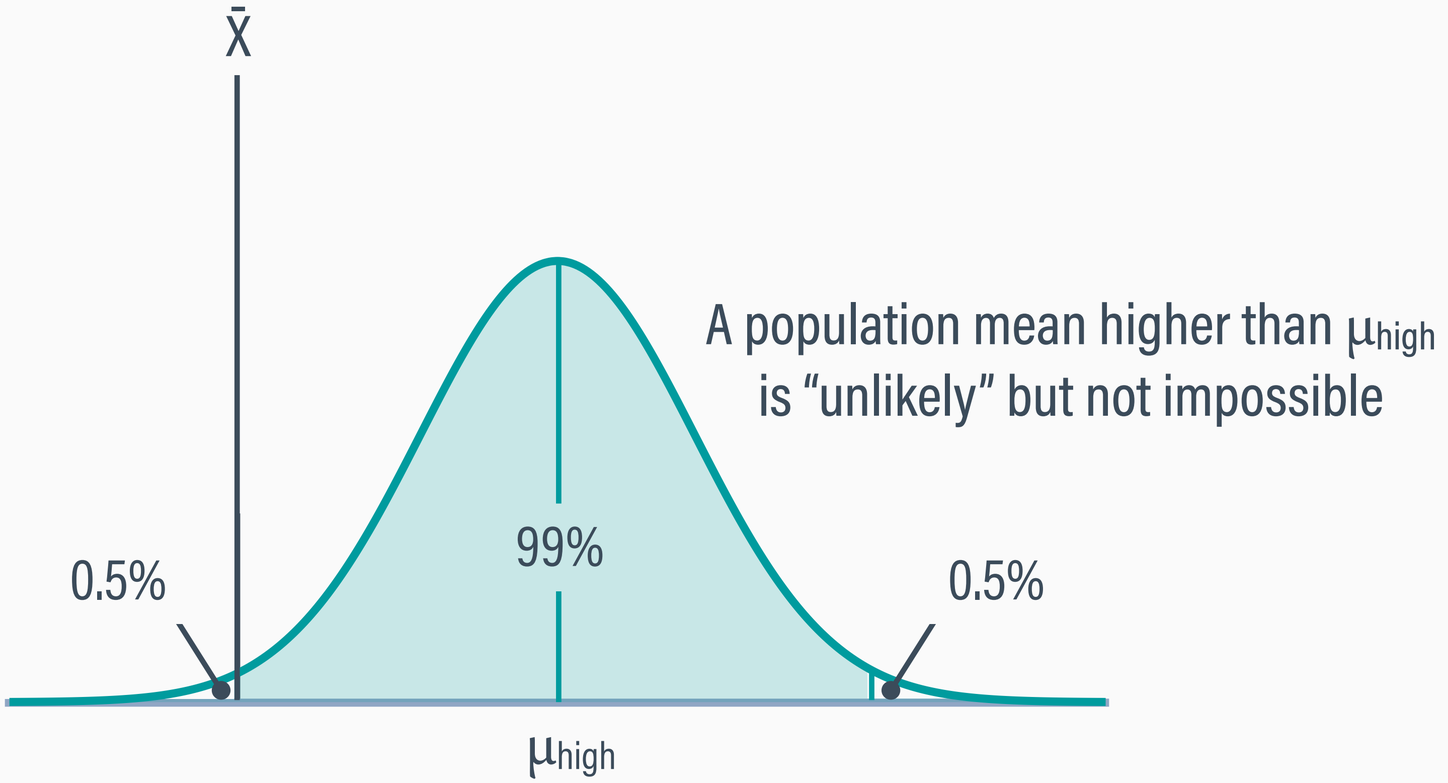
- 99% confidence implies that $\bar{X}$ sits at the upper boundary of the middle 99% (the 99.5th percentile) of all samples from a population with μ_{low}

A population mean lower than μ_{low} is "unlikely" but not impossible



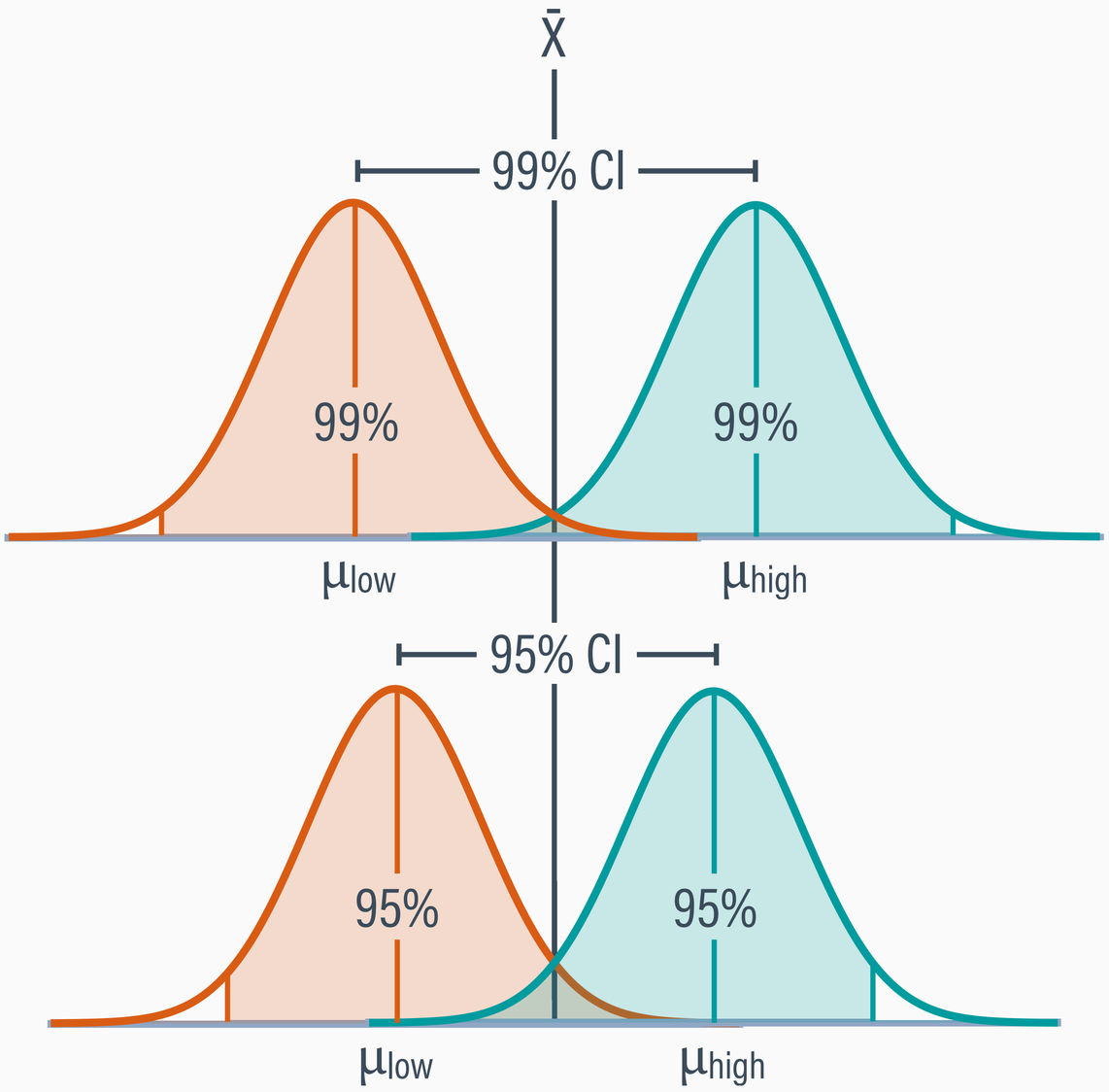
99% CONFIDENCE

- 99% confidence implies that $\bar{X}$ sits at the lower boundary of the middle 99% (the 0.5th percentile) of all samples from a population with μ_{high}



95% VS. 99% INTERVALS

- 95% confidence intervals are the norm
- 99% intervals are wider (encompass a broader range of plausible μ) because they allow an estimate to be more distant from μ

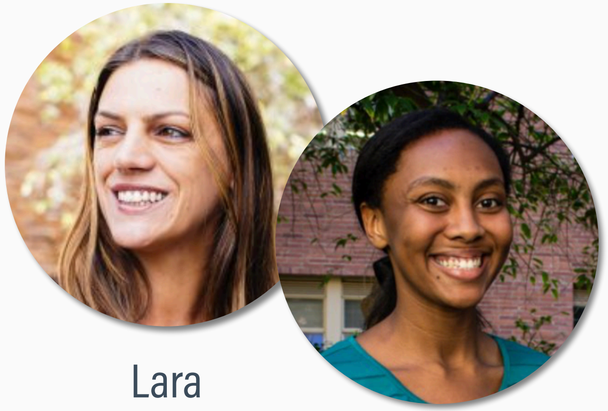


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SMOKING AND DRINKING CESSATION TRIAL

Pharmacological treatments that can concomitantly address cigarette smoking and heavy drinking stand to improve health care delivery for these highly prevalent co-occurring conditions. This superiority trial compared the combination of varenicline and naltrexone against varenicline alone for smoking cessation and drinking reduction among heavy-drinking smokers.



Lara Ray

ReJoyce Green

Ray, L.A., Green, R., Enders, C., et al. (2021). Efficacy of combining varenicline and naltrexone for smoking cessation and drinking reduction: A randomized clinical trial. *American Journal of Psychiatry*, *178*, 818–828.

KEY VARIABLES



Breath (alveolar) carbon monoxide

A measure of carbon monoxide in the lungs. Breath carbon monoxide is a biomarker of smoking behavior common in clinical trials. Higher scores reflect more frequent smoking.



Medication arm

Participants were randomly assigned to receive one of two meds: varenicline plus naltrexone or varenicline plus placebo pills

STEP 1: COMPUTE SAMPLE STATISTICS

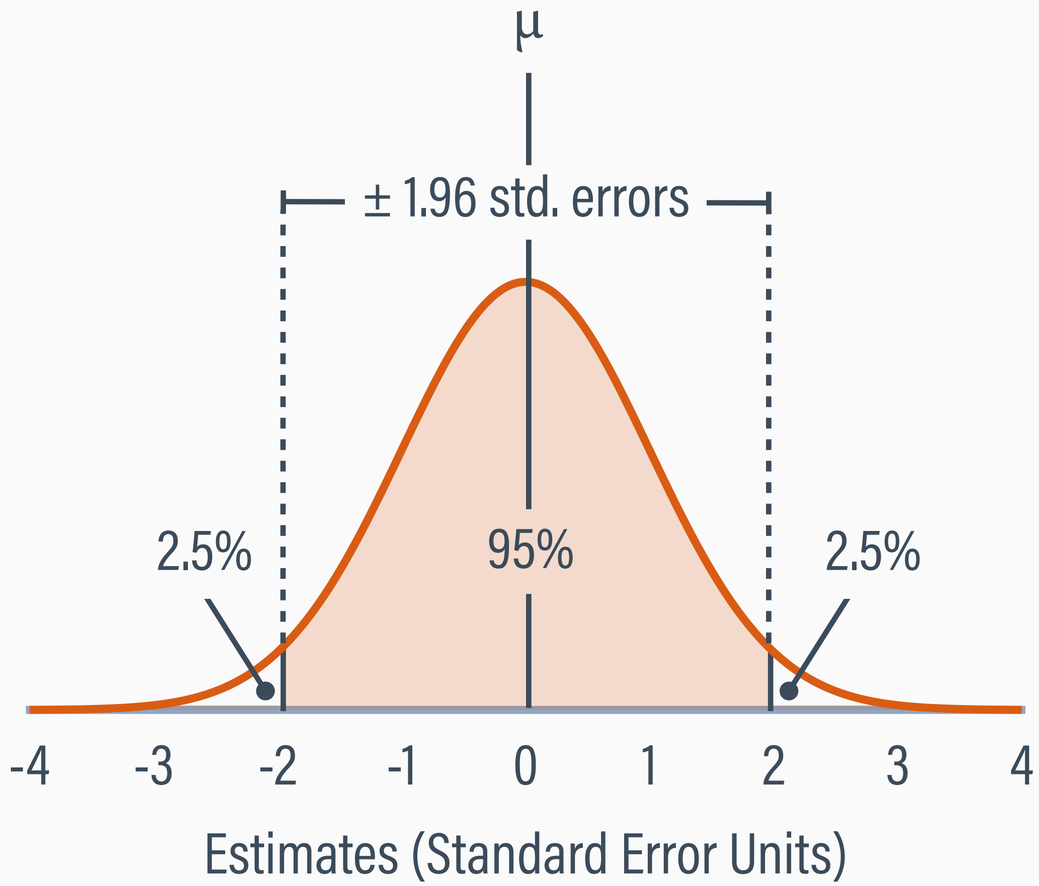
- The standard error is integral to computing confidence intervals
- The expected (average) sampling error for an estimate based on N = 165 is 0.47 breath CO points
- We expect this sample mean to be ±0.47 from the true mean

Mean ($\bar{X}$)	5.5
Std. Dev. (s)	5.96
N	165

$$s_{\bar{X}} = \frac{s}{\sqrt{N}} = \frac{5.96}{\sqrt{165}} = 0.47$$

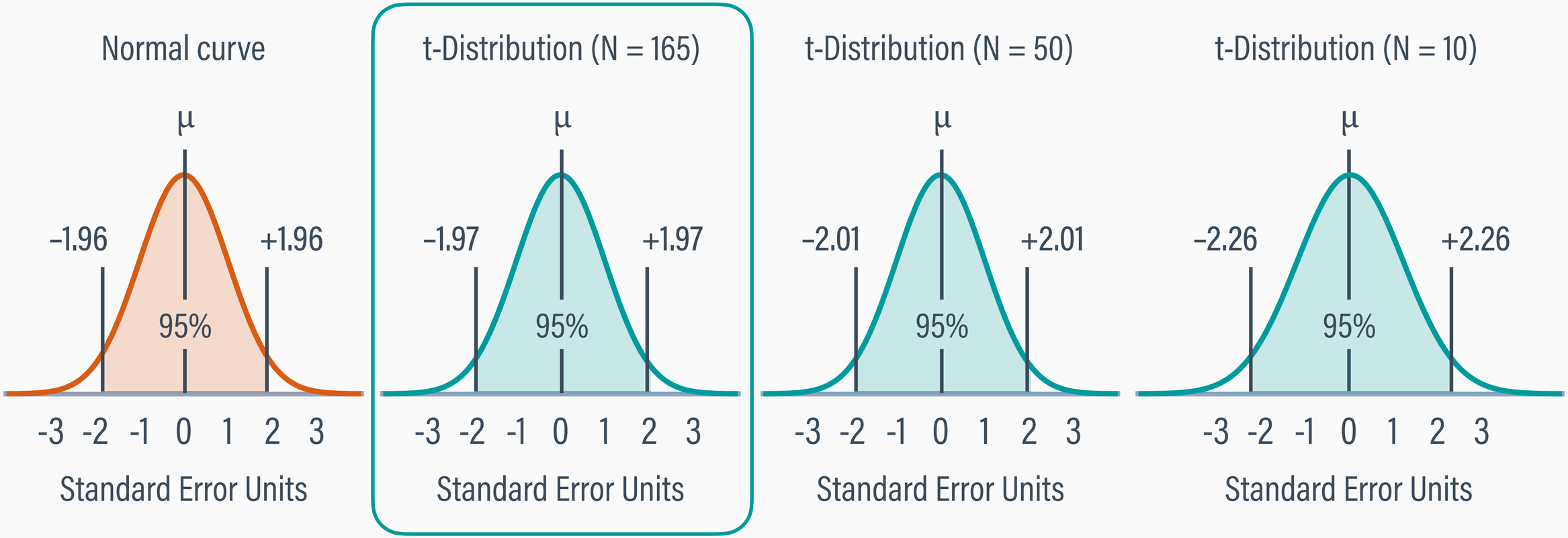
STEP 2: DETERMINE CRITICAL VALUES

- For 95% confidence, we need to determine the width of the 95% confidence range in standard error units (called a critical value)
- We can apply a normal curve approximation
- 95% of the means from large samples are within ± 1.96 standard errors of the true mean



T-DISTRIBUTION CRITICAL VALUES

- Software programs use a t-distribution where 95% critical values stretch out as the sample size decreases



STEP 3: SOLVE FOR INTERVAL LIMITS

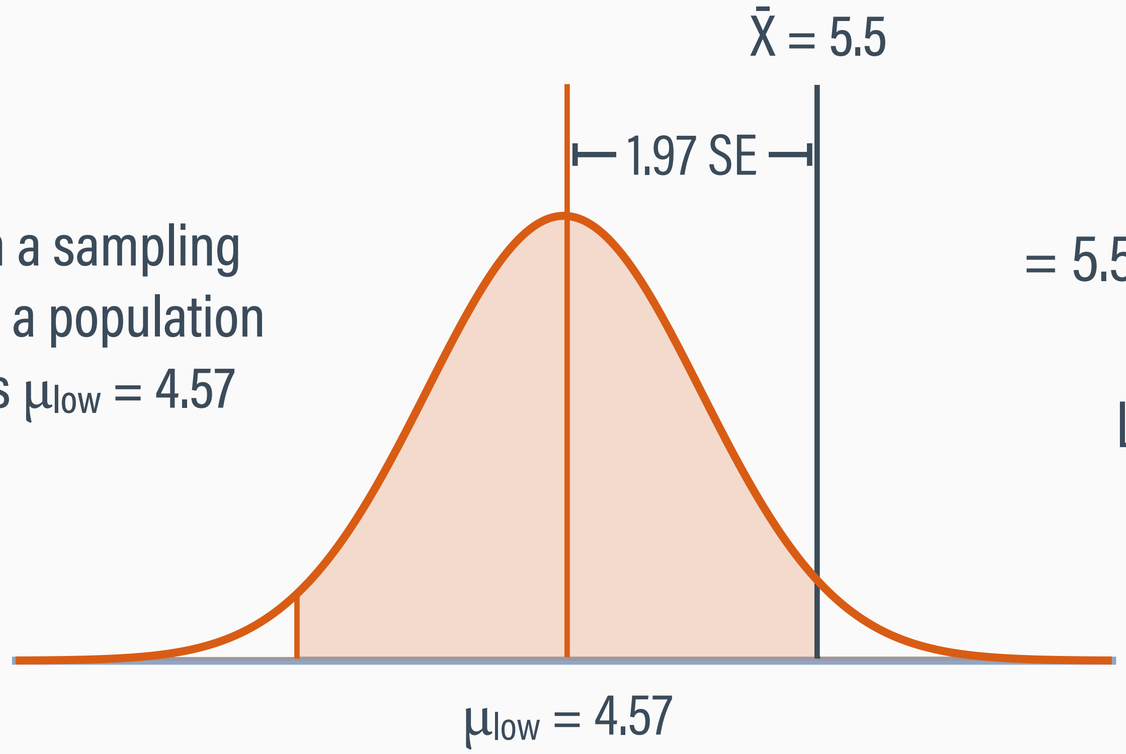
- We know that our estimate is 1.97 standard error units above (or below) the lowest (or highest) plausible values of μ
- With the standard error, we can easily solve for μ_{low} and μ_{high}
- $CI_{95\%} = \bar{X} \pm (\text{critical value} \times \text{standard error})$

LOWEST PLAUSIBLE POPULATION MEAN

- The lowest plausible value of μ lies 1.97 standard error units ($\approx 1.97 \times 0.47 = 0.93$ breath CO points) below the sample mean

Polls call this the margin of error

$\bar{X}$ could be from a sampling distribution with a population mean as low as $\mu_{low} = 4.57$



$$CI_{95\%} = \bar{X} \pm (1.97 \times \text{standard error})$$

$$= 5.5 \pm (1.97 \times 0.47) = [\text{Lower}_{95\%}, \text{Upper}_{95\%}]$$

$$\text{Lower}_{95\%} = 5.5 - (1.97 \times 0.47) = 4.57$$

HIGHEST PLAUSIBLE POPULATION MEAN

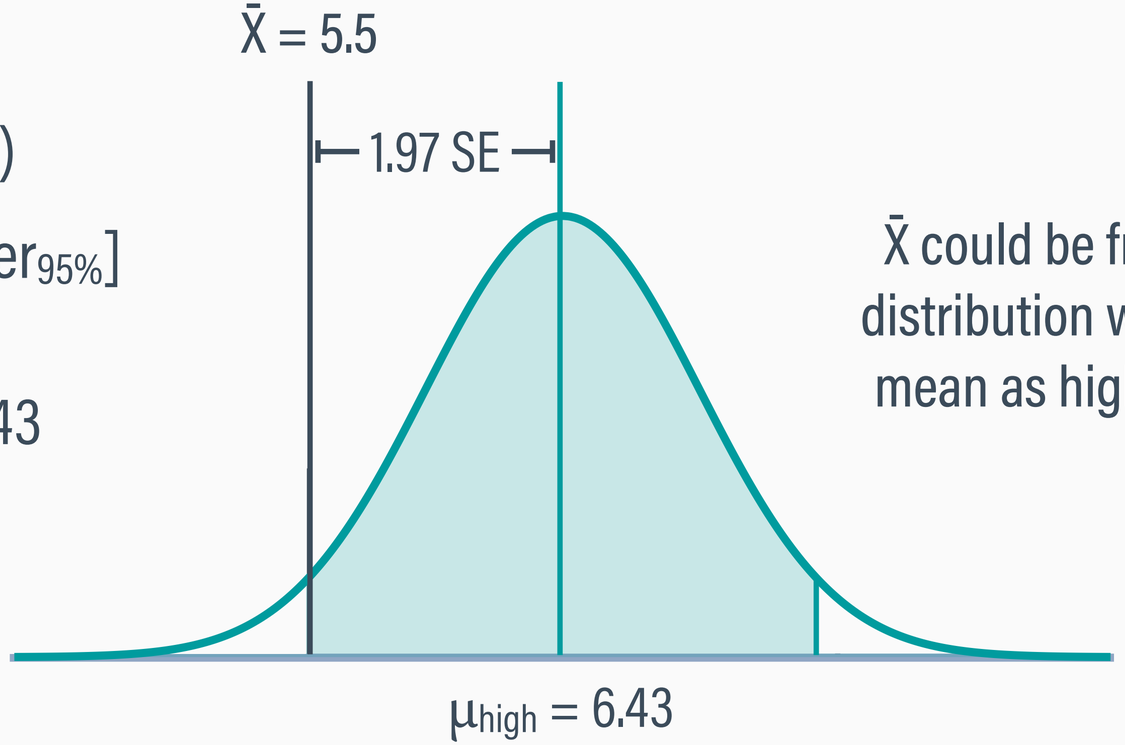
- The highest plausible value of μ lies 1.97 standard error units ($\approx 1.97 \times 0.47 = 0.93$ breath CO points) above the sample mean

Polls call this the margin of error

$$CI_{95\%} = \bar{X} \pm (1.97 \times \text{standard error})$$

$$= 5.5 \pm (1.97 \times 0.47) = [\text{Lower}_{95\%}, \text{Upper}_{95\%}]$$

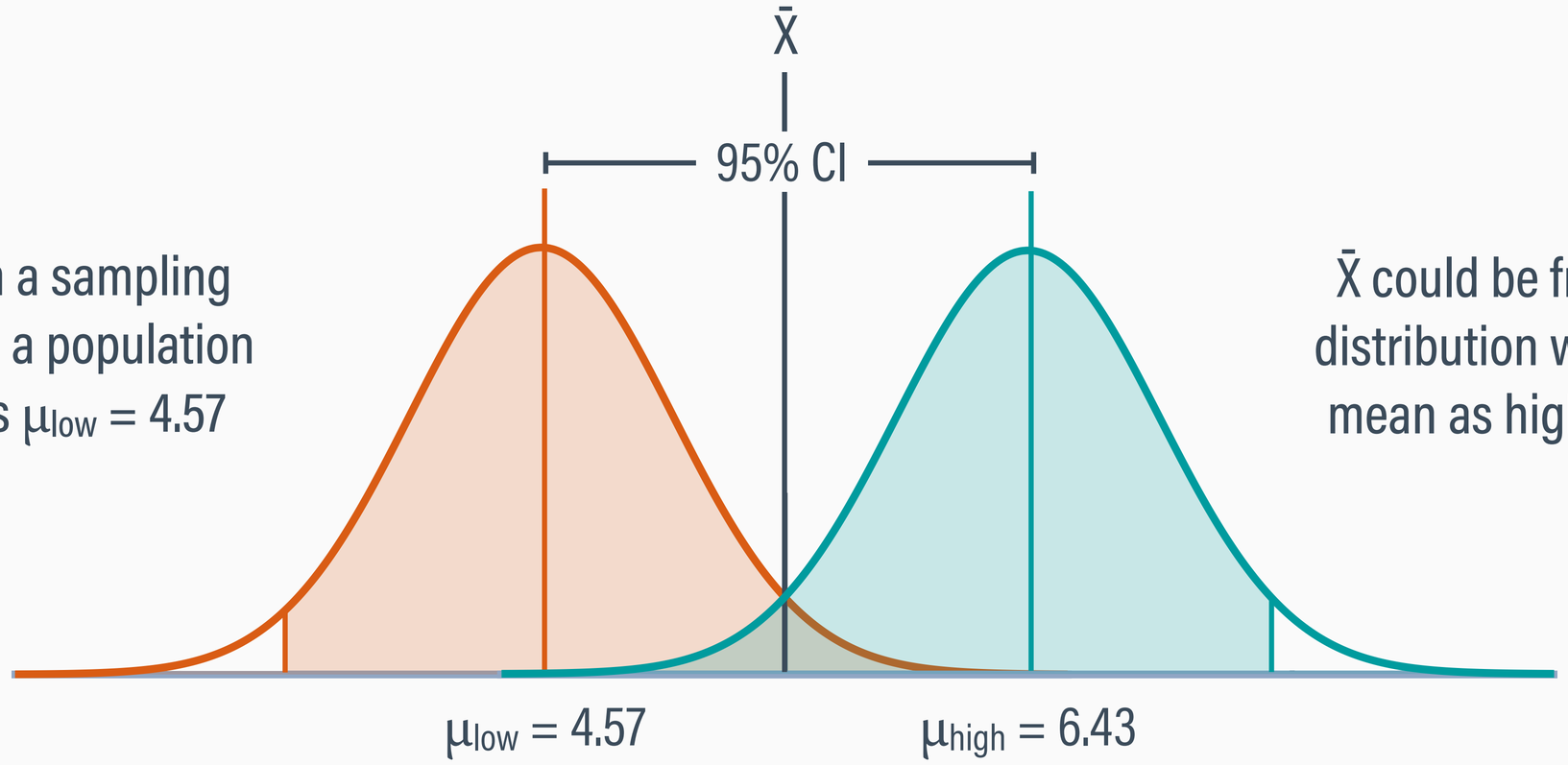
$$\text{Upper}_{95\%} = 5.5 + (1.97 \times 0.47) = 6.43$$



$\bar{X}$ could be from a sampling distribution with a population mean as high as $\mu_{high} = 6.43$

95% CONFIDENCE INTERVAL

- $CI_{95\%}$ = [4.57, 6.43] gives the two most extreme values of the population mean that could have reasonably produced the data



R OUTPUT

```
    Mean     Lower     Upper 
5.527273 4.611307 6.443238 
```



The clinical trial produced a mean and 95% confidence interval of $\bar{X} = 5.5$ and $CI_{95\%} = [4.57, 6.43]$. It is widely believed that breath CO levels of 5 reflect successful outcomes in clinical trials. In small groups of two or three, discuss whether this sample of of $N = 165$ participants could have reasonably originated from a population with a true mean of $\mu = 5$ (i.e., a population where the trial was deemed a success).

HYPOTHESIS TESTING WITH 95% INTERVALS

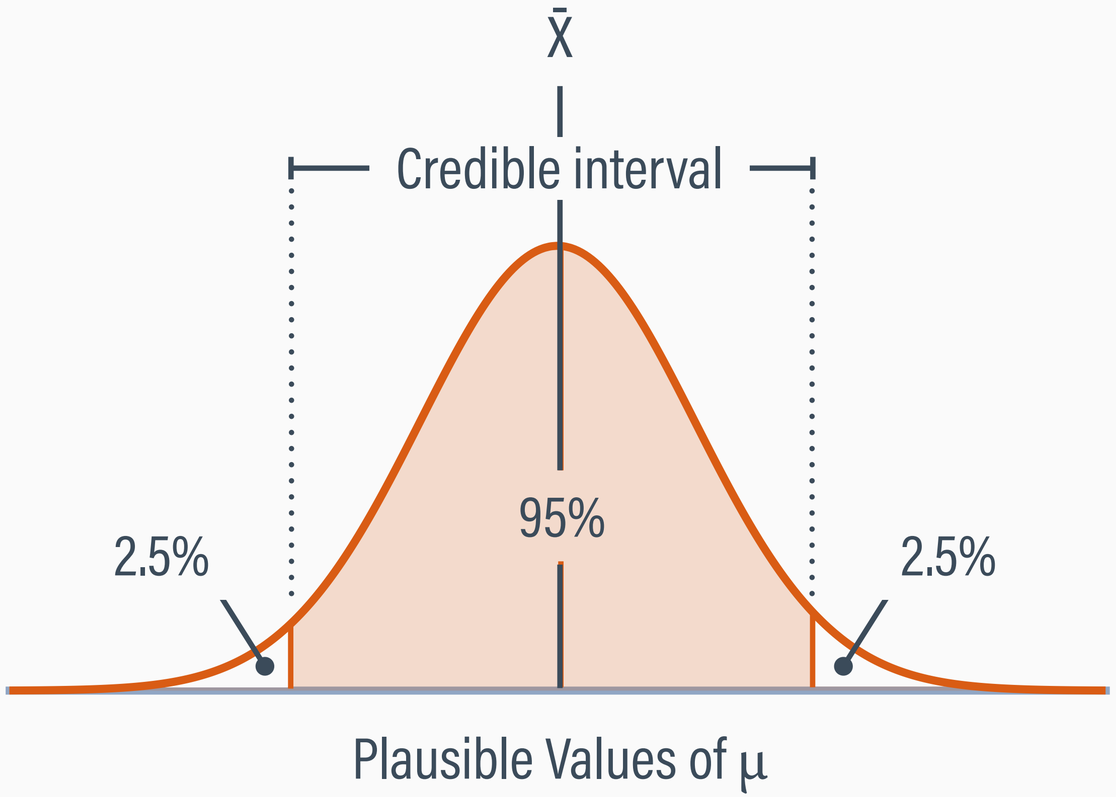
- The clinical trial produced $\bar{X} = 5.5$ and $CI_{95\%} = [4.57, 6.43]$
- It is widely believed that breath CO levels of 5 reflect successful outcomes in clinical trials
- The data support the hypothesis that the true mean is $\mu = 5$ because that value is within the interval (a population with $\mu = 5$ could have plausibly produced these data)

ALTERNATE INTERPRETATIONS

- The “extreme population” definition of confidence intervals is from Cumming and Finch (2001)
- In significance testing, the CI defines a set of null hypotheses that the data could not reject (Cox and Hinkley, 1974)
- The CI is a range of values that are plausible for μ, and values outside it are relatively implausible (Cumming, 2014)— this is dangerously close to a Bayesian interpretation

BAYESIAN INTERPRETATION

- Many values of the population μ could produce our data (no hypothetical samples)
- There is a distribution of plausible μ values, and $\bar{X}$ is the most likely value, located at the center of the parameter (posterior) distribution
- A 95% credible interval simply defines a range of plausible population μ values



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MEANING OF 95% CONFIDENCE

- Researchers often say things like, “we are 95% confident that the true mean falls somewhere in the interval $CI_{95\%} = [4.57, 6.43]$” or “there is a 95% probability that the true mean falls somewhere in the interval $CI_{95\%} = [4.57, 6.43]$”

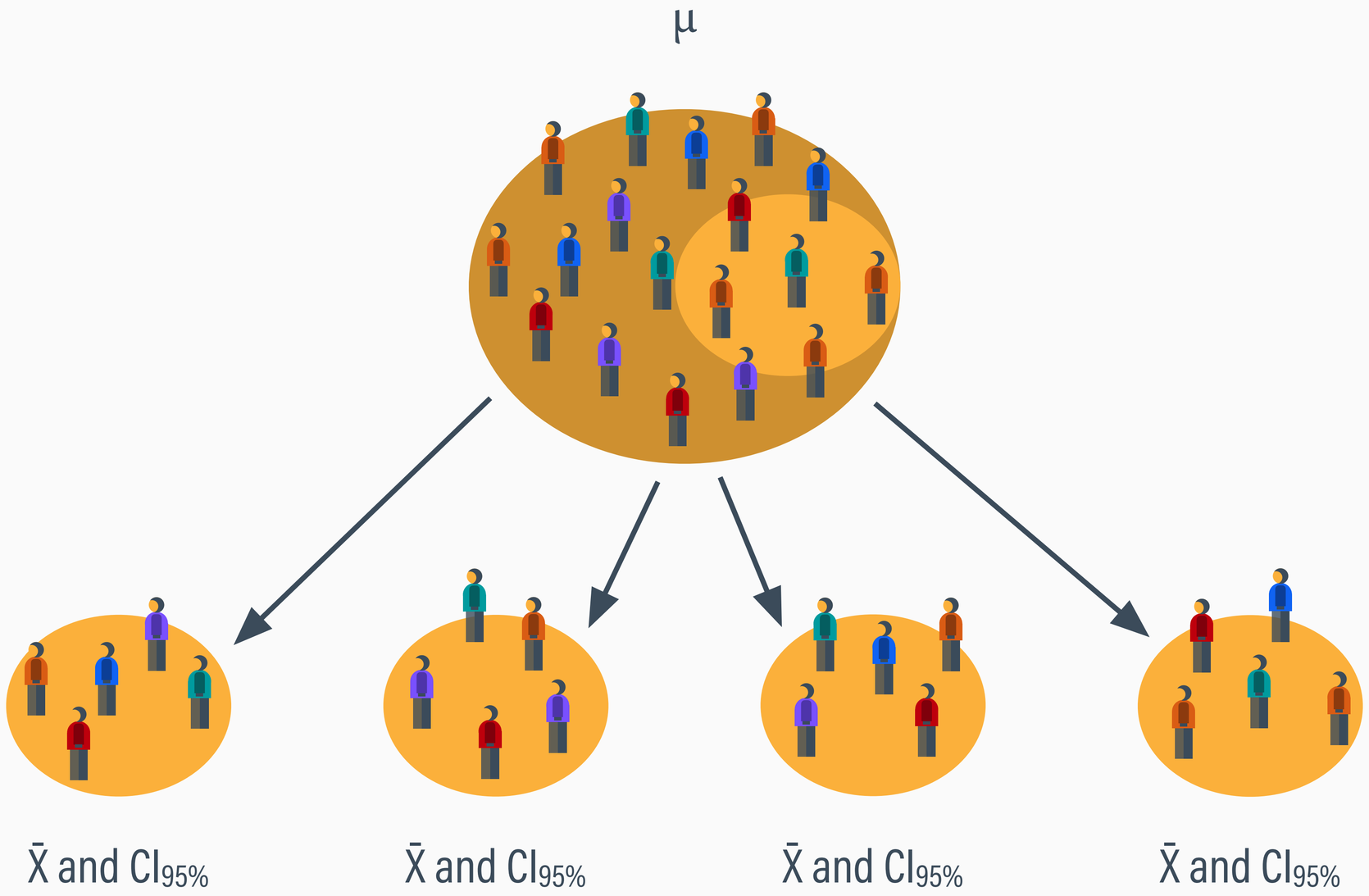
- This language is a bit sloppy and imprecise because it does not convey what is meant by 95% confident or 95% probable

FREQUENTIST PARADIGM REVISITED

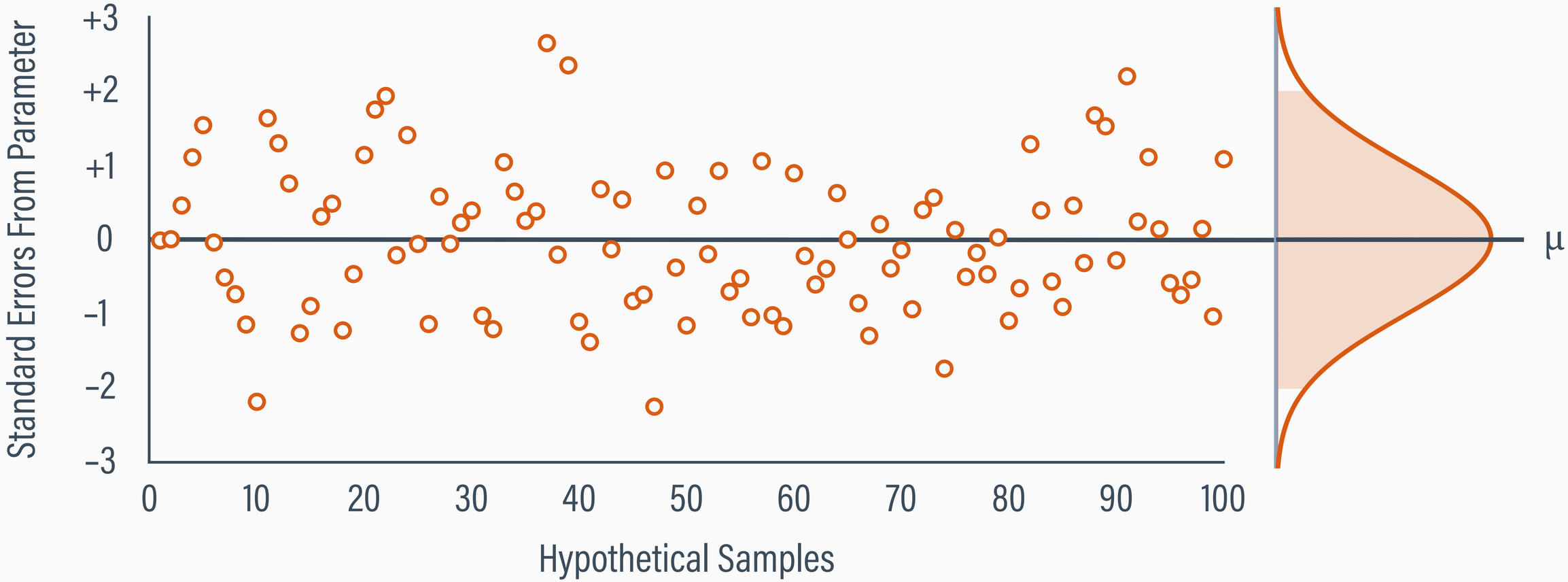
- The frequentist paradigm is defined by the idea that there is one population with unknown parameters (e.g., μ and σ)
- We imagine numerous hypothetical samples of size N from the population, each with its own unique estimates and 95% CIs
- The population-level statistics (parameters) are locked in at a single set of values, whereas the sample-level statistics (estimates) vary across different data sets

FREQUENTIST FRAMEWORK

One population with unknown parameters. Estimates and CIs vary across different hypothetical samples.



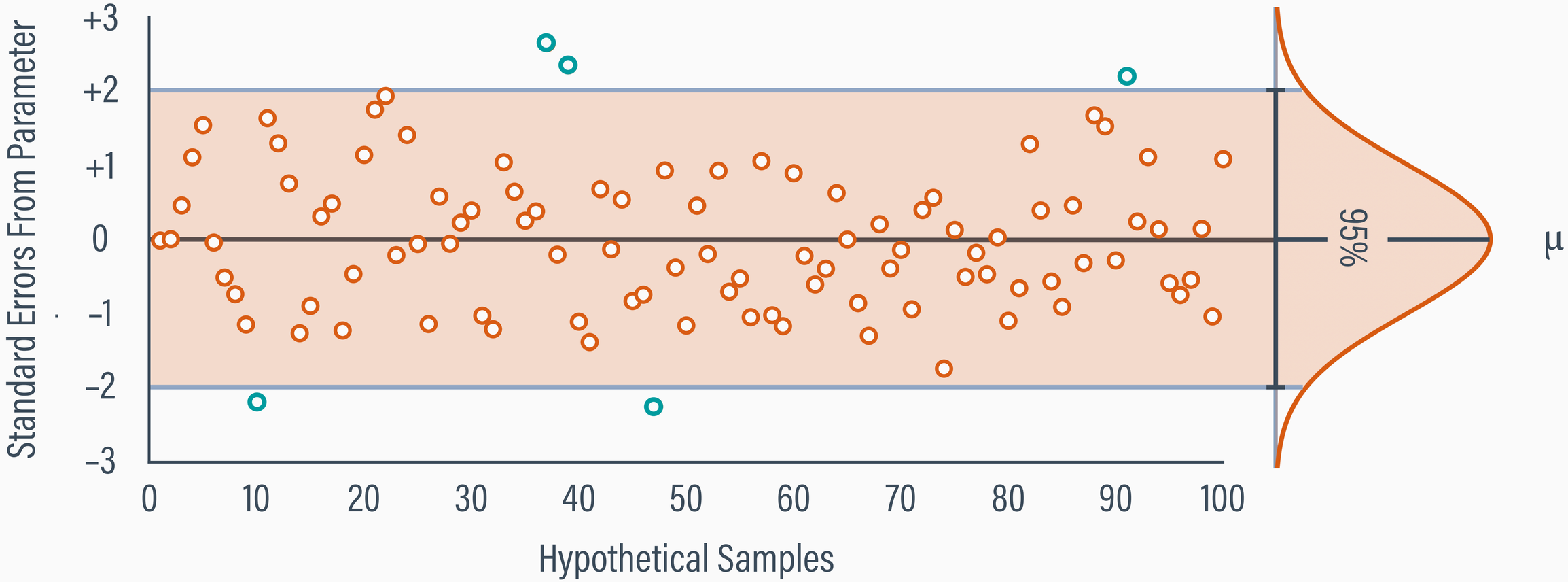
ESTIMATES FROM 100 RANDOM SAMPLES



95% RANGE AROUND TRUE MEAN

= estimate outside middle 95% range

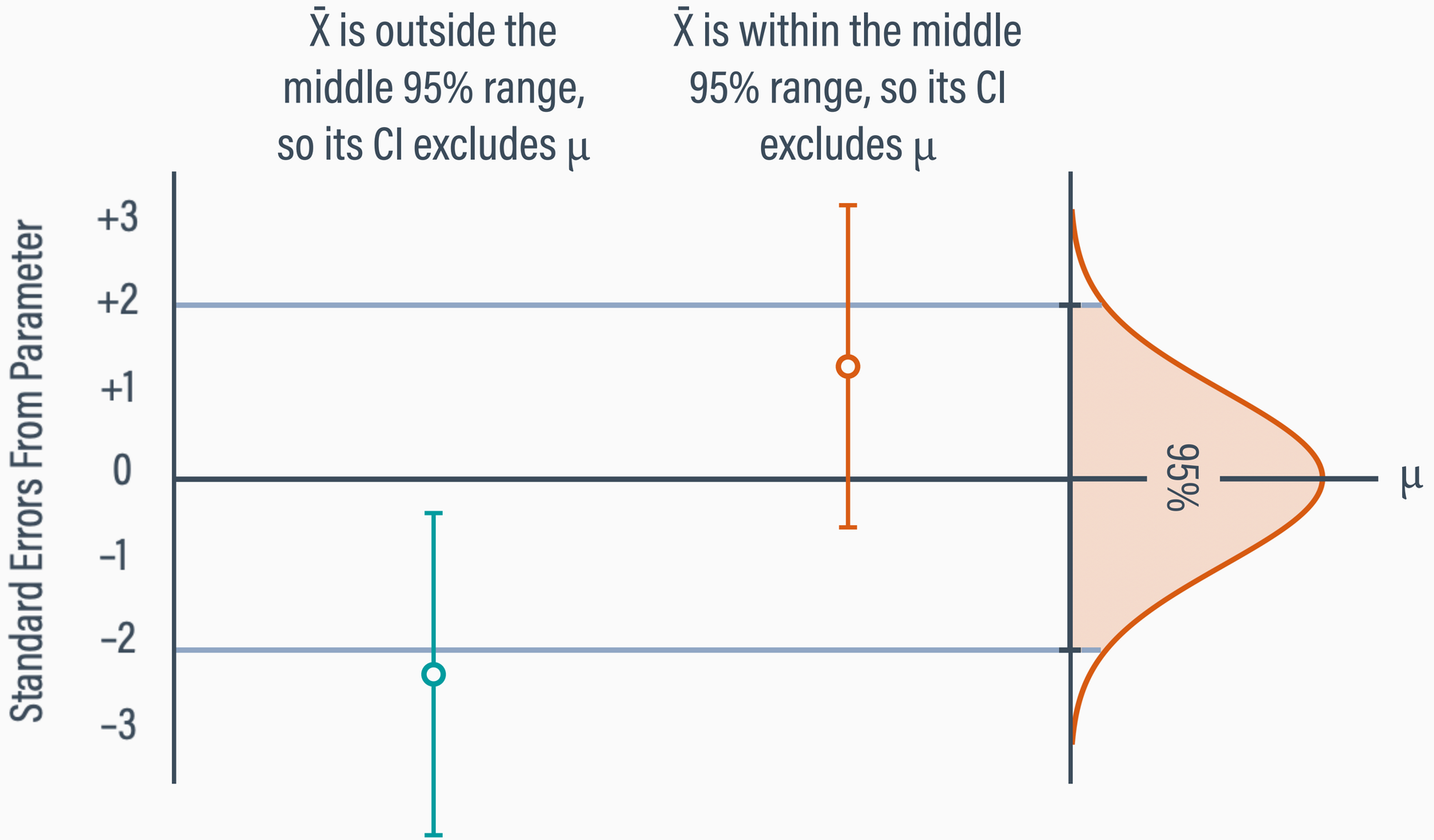
= estimate within middle 95% range



CONFIDENCE = LONG RUN PROBABILITY

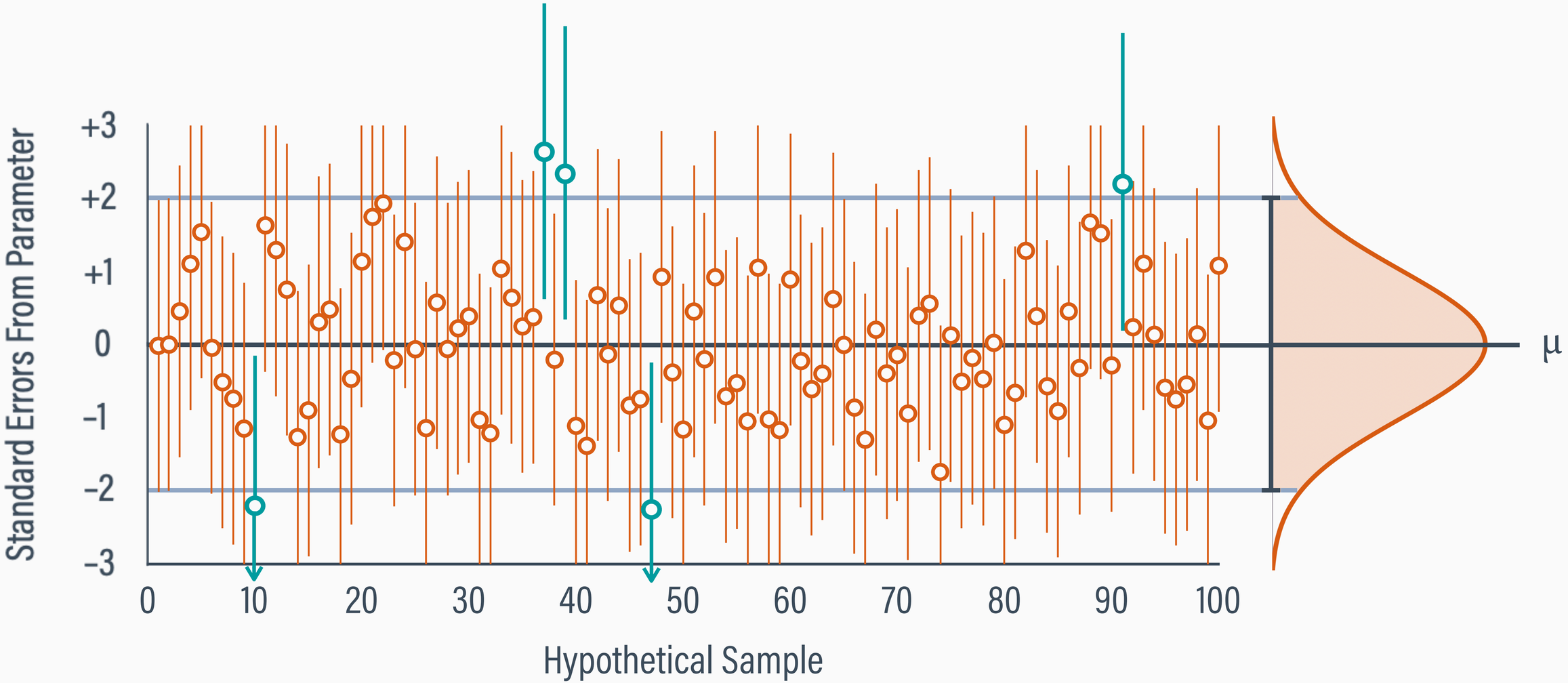
- Each sample has a unique mean and confidence interval
- In any given sample, the interval either contains the true population mean or it does not (the probability is either 0 or 1)
- Confidence and probability are concepts that unfold over many hypothetical random samples from a population
- Across many samples, there is a 95% probability that a confidence interval contains the true population statistic

INTERVALS FROM SINGLE SAMPLES



95% OF INTERVALS CONTAIN THE TRUE MEAN

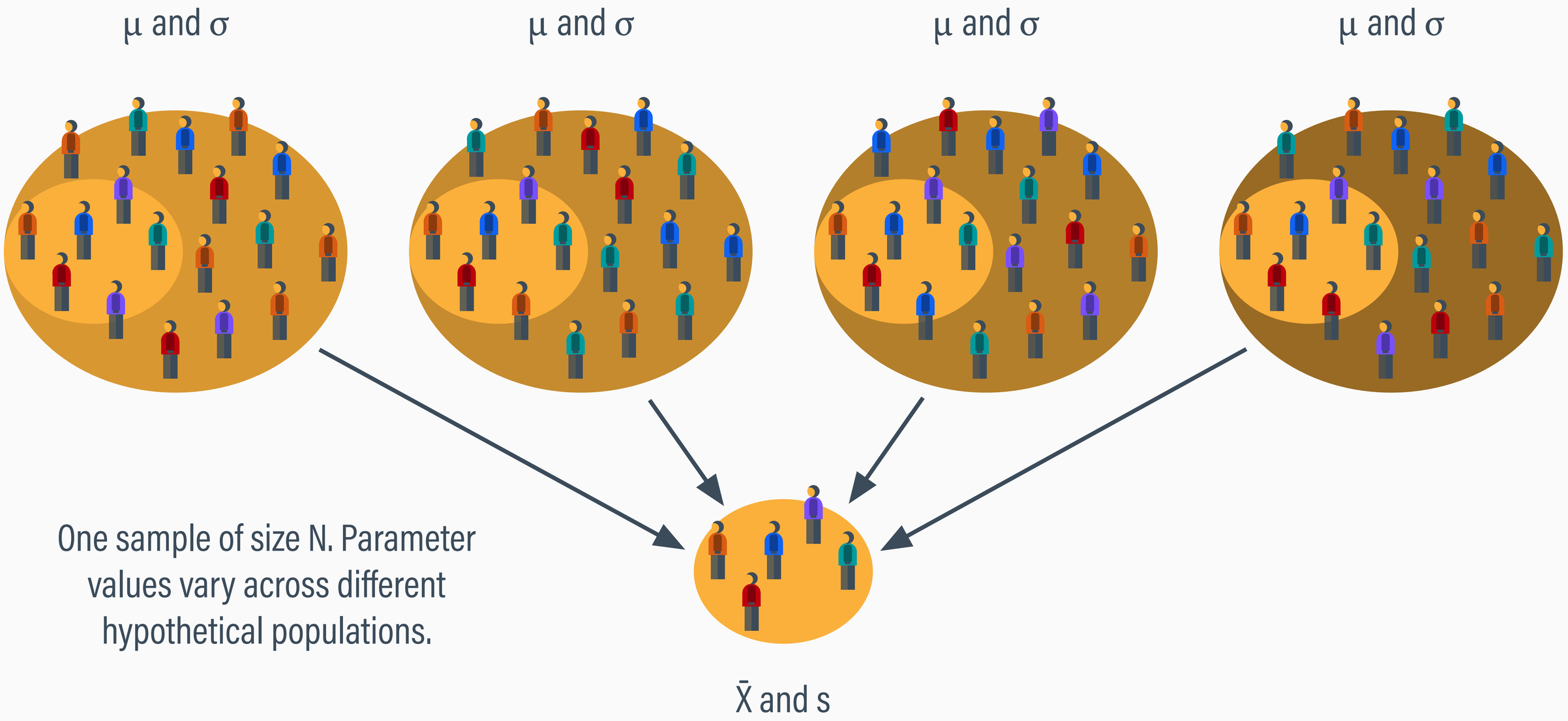
On average, the 95 out of every 100 samples should give a CI that includes the true population statistic





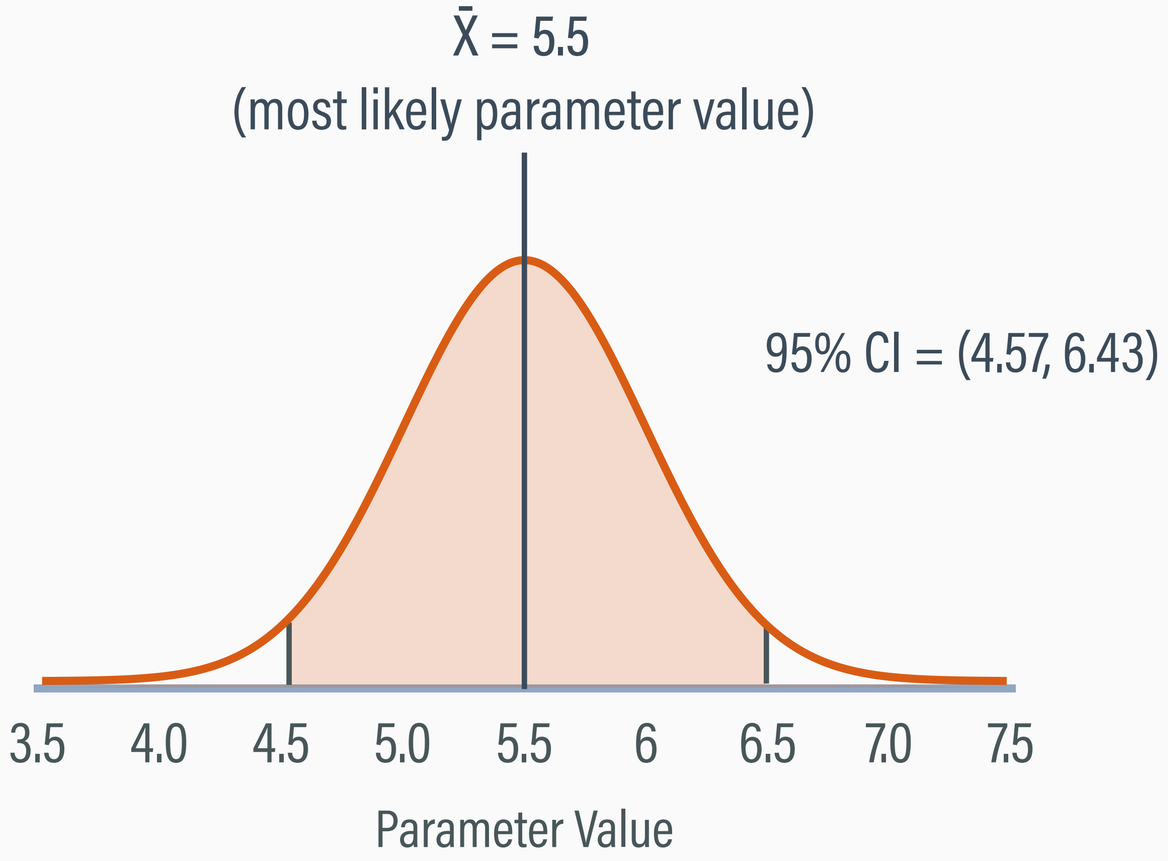
The clinical trial produced a mean and 95% confidence interval of $\bar{X} = 5.5$ and $CI_{95\%} = [4.57, 6.43]$. Consider the following statement: “There is a 95% probability (we are 95% confident) that the true mean ranges between 4.57 and 6.43”. In small groups of two or three, discuss why this statement is incorrect (or misleading, at best).

BAYESIAN FRAMEWORK



BAYESIAN 95% CREDIBLE INTERVAL

- A distribution (called a posterior) describes the plausible parameter values for the data
- $CI_{95\%} = [4.57, 6.43]$ includes 95% of the population means that could have reasonably produced the data
- There is a 95% probability that the parameter falls between 4.57 and 6.43



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COMPARATIVE RESEARCH QUESTIONS

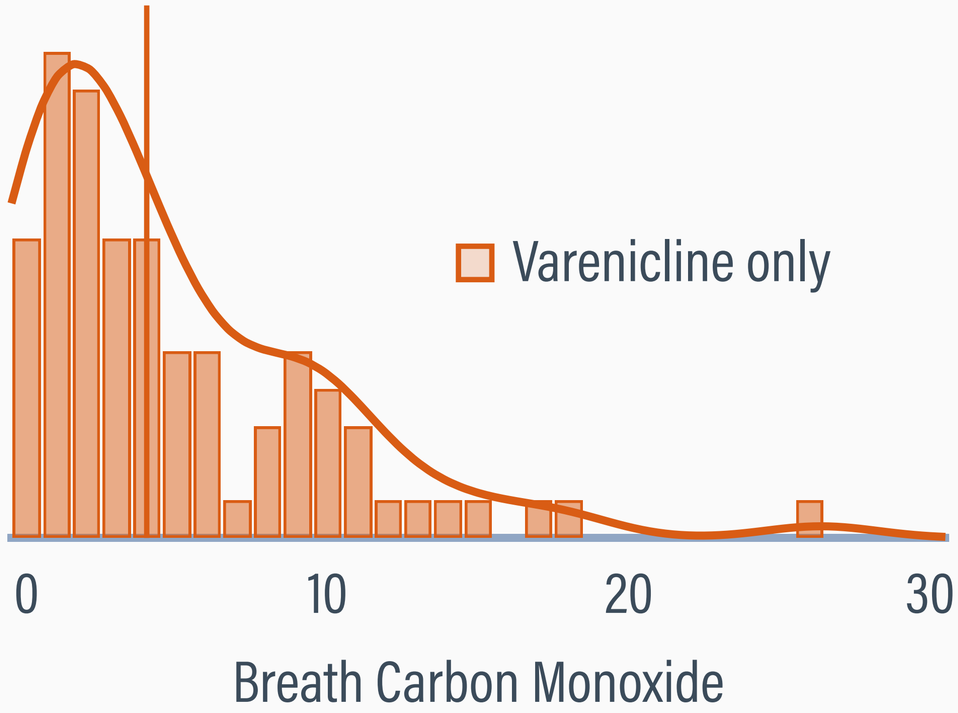
- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Do participants in the two treatment groups differ in their smoking levels?
- We can answer this question by visually comparing group-specific 95% confidence intervals

GROUP MEANS AND INTERVALS

$\bar{X} = 5.02$

$s = 4.88$

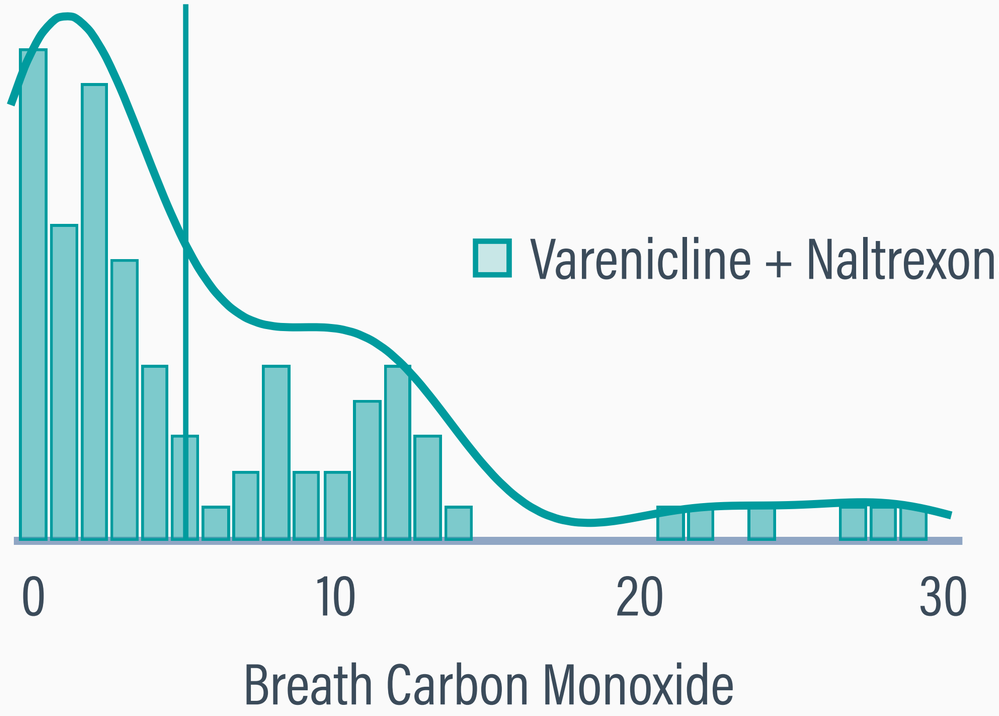
$CI_{95\%} = [3.95, 6.09]$



$\bar{X} = 6.02$

$s = 6.86$

$CI_{95\%} = [4.52, 7.52]$

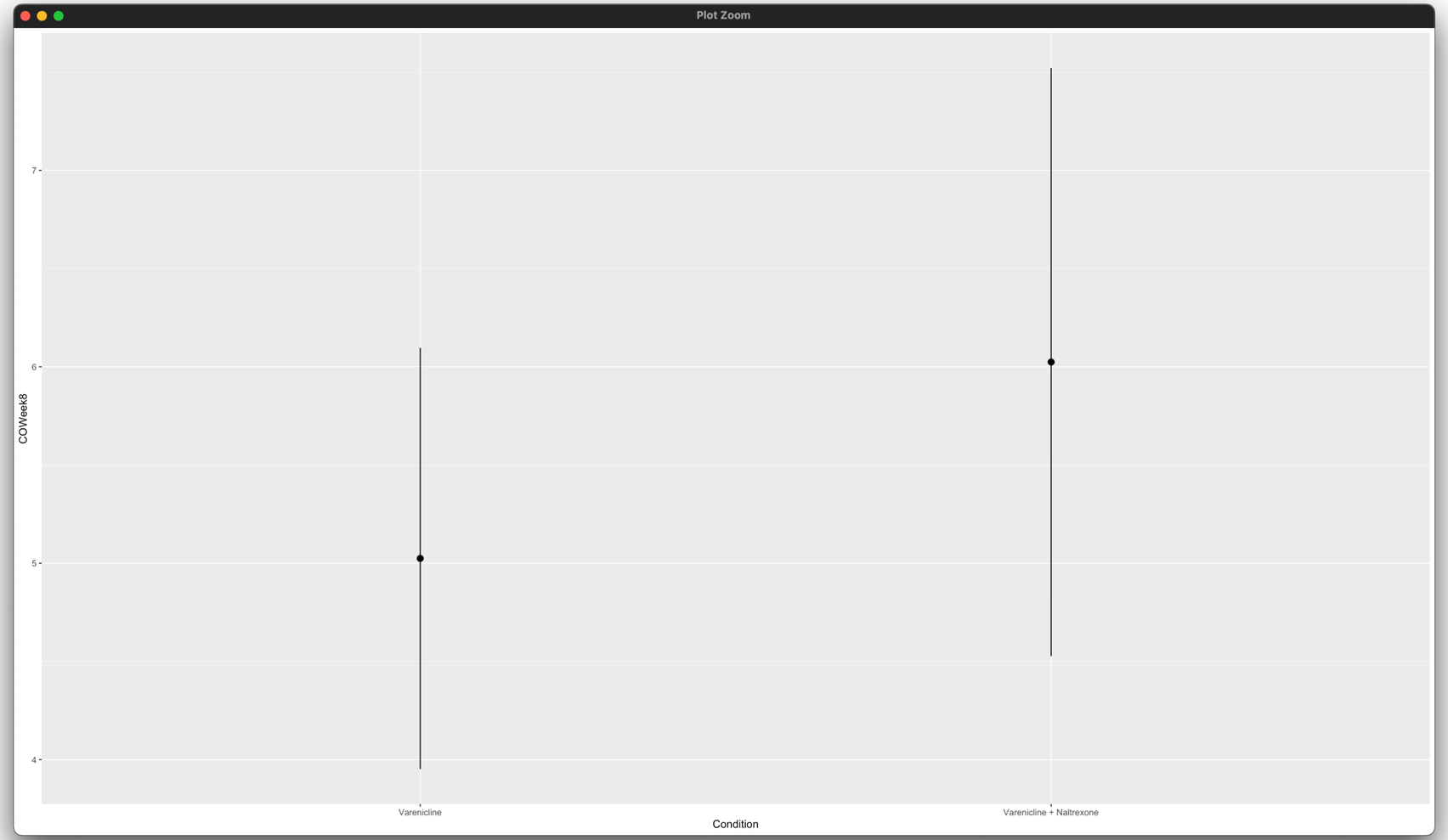


R OUTPUT

```
$Varenicline
    Mean    Lower    Upper 
5.024390 3.952419 6.096362 

$`Varenicline + Naltrexone`
    Mean    Lower    Upper 
6.024096 4.527132 7.521061 
```

R OUTPUT

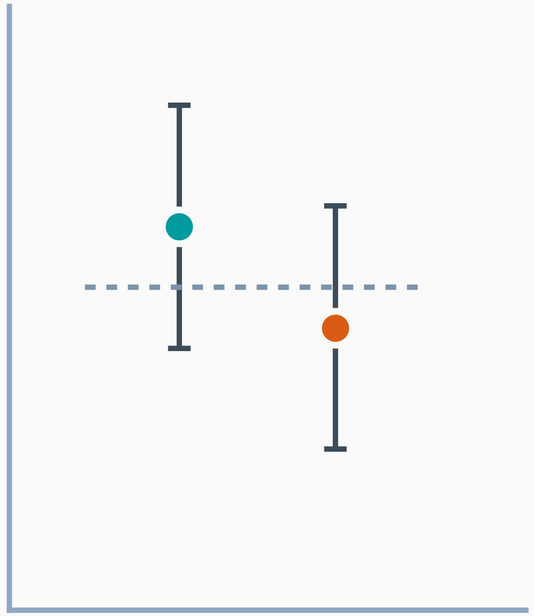

Plot Zoom
7
6
5
4
COWeek8
Varenicline
Varenicline + Naltrexone
Condition

INFERENCE BY EYE

- The amount of confidence interval overlap provides information about whether the difference between two groups is “statistically significant”
- If error bars overlap by less than 50%, the group means differ beyond chance expectations
- The difference is “significant at $p < .05$”

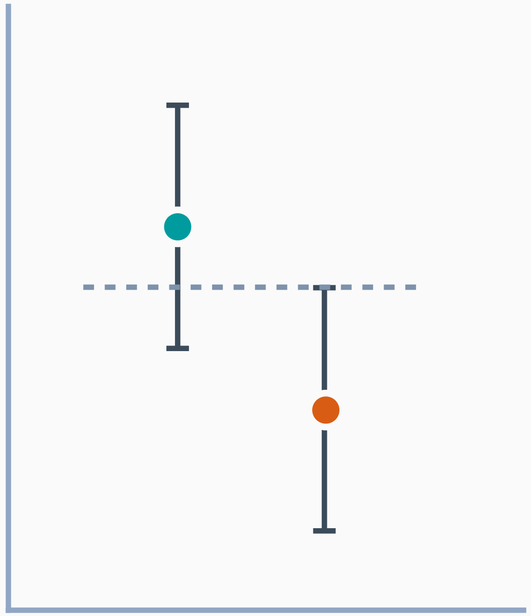
INFERENCE BY EYE

Nonsignificant
mean difference ($p > .05$)



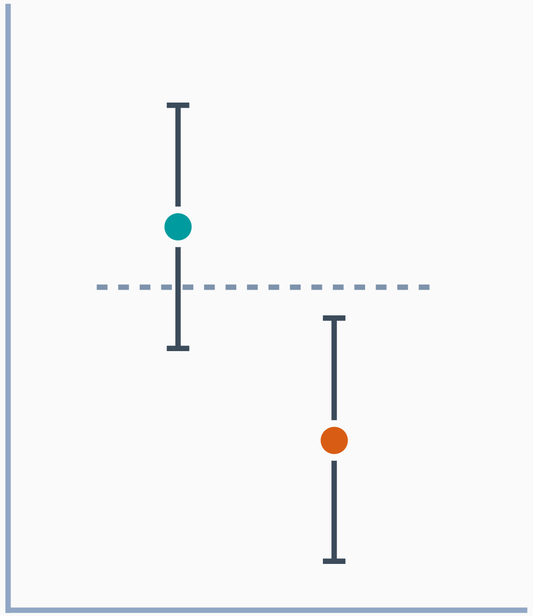
More than 50%
overlap in 95% CIs

Indeterminate
mean difference ($p = .05$)



Exactly 50%
overlap in 95% CIs

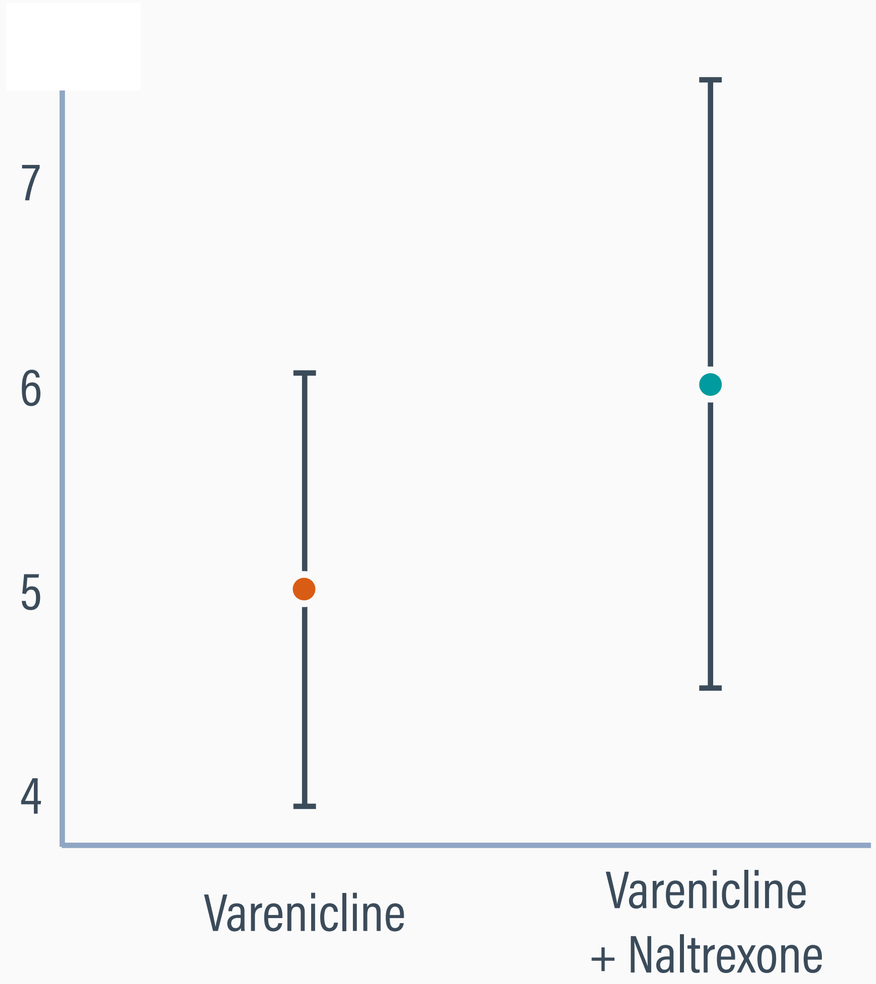
Significant
mean difference ($p < .05$)



Less than 50%
overlap in 95% CIs

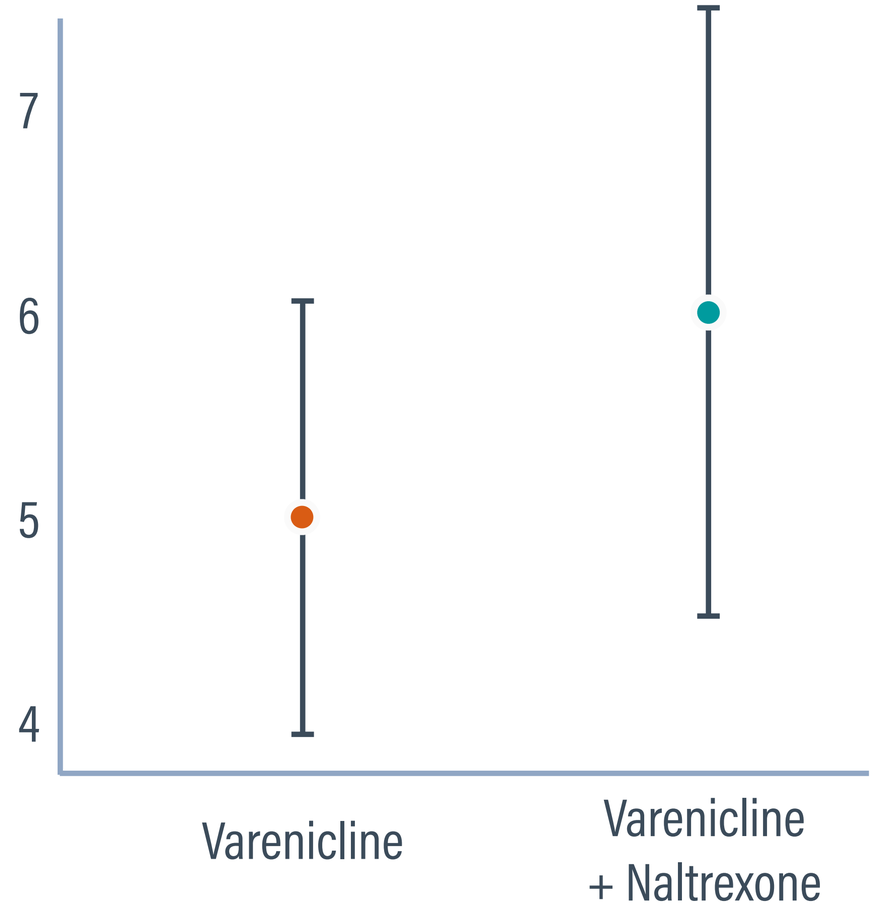


One of the main goals of the research study is to determine whether the treatment groups differ. In small groups of two or three, apply the inference by eye rule. Do the groups exhibit a statistically meaningful difference?





The dual medication group's 95% interval is noticeably wider, despite the groups being the same size. In small groups of two or three, discuss potential causes for this difference. Hint: look at the descriptive statistics.



OUTLINE

1. Quick review
2. Overview of confidence intervals
3. Constructing 95% confidence intervals
4. Properties of confidence intervals
5. Inference by eye
6. Study questions

STUDY QUESTIONS

- The study questions continue using the clinical trial research scenario introduced in the lecture, including new variables
- The questions illustrate the kinds of statistical reasoning the exams will require, including interpreting evidence, evaluating conclusions, and connecting graphical and numerical results
- Midterm exam questions will require you to apply these same reasoning skills to new research scenarios

R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Male*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37

R OUTPUT

```
> CI(ClinicalTrial$DrinksWeek8, ci = 0.95)

   upper     mean    lower 
3.642848 3.230303 2.817758 
```

STUDY QUESTIONS (1)

At the 8-week follow-up, participants reported the number of alcoholic drinks consumed during the past week (DrinksWeek8). The following study questions refer to this variable.

1. Write an explanation of the standard error of 0.21 to an undergraduate research assistant. Identify what quantity varies across repeated samples and interpret the numerical value in the context of this study.

2. Using the “extreme populations” interpretation presented in the lecture, explain what the lower and upper 95% confidence limits represent. Include both numerical limits and specify which quantities are held fixed and which are varied in this interpretation.

3. Consider the statement: “There is a 95% probability that the interval from 2.82 to 3.64 contains the true population mean number of drinks.” Explain what “95% confident” means in the frequentist paradigm, specifying what is fixed and what varies.

STUDY QUESTIONS (2)

4. Compute the width of the confidence interval.

5. Suppose the sample size doubled from 165 to 330 while the sample standard deviation and confidence level (95%) remained unchanged. What would happen to the width of the confidence interval? Why?

6. Keeping the sample size and standard deviation unchanged, what would happen to the interval's width if the researchers requested 99% confidence rather than 95%? What if they requested 90% confidence? Explain the reasoning behind both answers.

STUDY QUESTIONS (3)

7. Suppose the researchers want to determine whether the population mean is 3 drinks per week. That is, they want to know whether this sample mean could have originated from a population where the true mean is 3 drinks. Explain what the interval allows you to conclude about this question and whether it establishes that the true population mean could equal 3.

8. Suppose the researchers want to determine whether the population mean is 5 drinks per week. That is, they want to know whether this sample mean could have originated from a population where the true mean is 5 drinks. Explain what the interval allows you to conclude about this question and whether it establishes that the true population mean could equal 5.

R OUTPUT

```
> describeBy(DrinksWeek8 ~ Condition, data = ClinicalTrial)

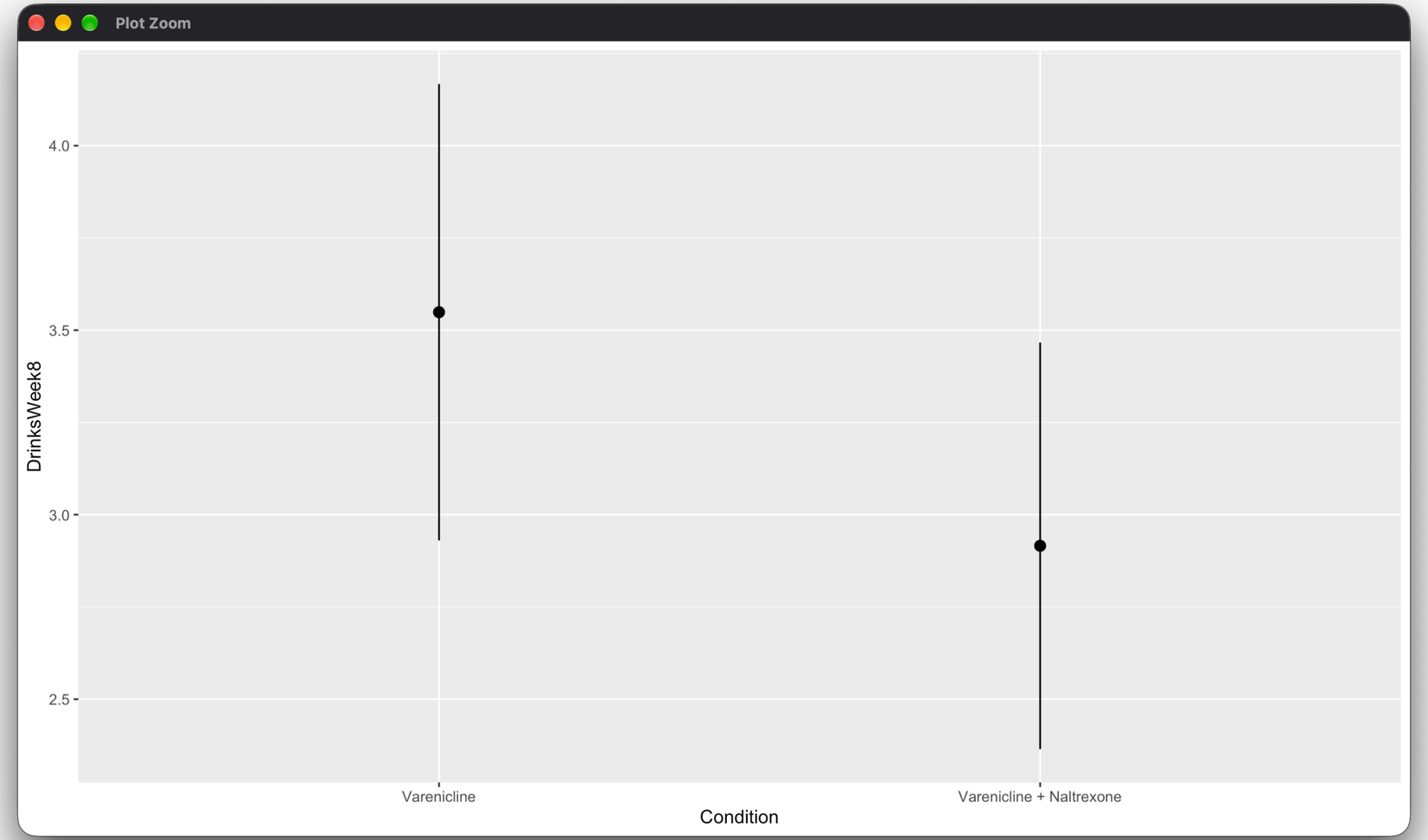

 Descriptive statistics by group
Condition: Varenicline
            vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
DrinksWeek8    1 82 3.55 2.82      3    3.26 2.97   0  13    13 1.01     0.93 0.31
------------------------------------------------------------------------------
Condition: Varenicline + Naltrexone
            vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
DrinksWeek8    1 83 2.92 2.52      2     2.7 2.97   0  11    11 0.64    -0.28 0.28



> group.CI(DrinksWeek8 ~ Condition, data = ClinicalTrial, ci = 0.95)


                 Condition DrinksWeek8.upper DrinksWeek8.mean DrinksWeek8.lower
1              Varenicline          4.167517         3.548780          2.930044
2 Varenicline + Naltrexone          3.466805         2.915663          2.364520
```

R OUTPUT


Plot Zoom
4.0
3.5
3.0
2.5
DrinksWeek8
Varenicline
Varenicline + Naltrexone
Condition

STUDY QUESTIONS (4)

9. The 95% confidence interval for mean drinking is wider in the varenicline group (4.17 – 2.93 = 1.24 drinks) than in the varenicline-plus-naltrexone group (3.47 – 2.36 = 1.11 drinks). Using the group-specific descriptive statistics, explain why the widths differ. Support your explanation with specific numerical values.

10. Do the results suggest that the two medication groups differ in the population mean number of drinks consumed at the 8-week follow-up? Apply the “inference by eye” rule for two independent groups to the graph. Describe the overlap of the confidence intervals and explain how it relates to the rule’s criterion. State your conclusion, distinguishing the observed direction of the sample mean difference from the evidence for a population difference.